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Supplementary information

Deep tissue multi-photon imaging using adaptive optics with direct focus sensing and shaping

In the format provided by the authors and unedited

Principle of α-FSS. This analysis of the operating principle takes three-photon excited fluorescence as an example and can be easily derived for other nonlinear effects. In the case of three-photon excitation, the fluorescence signal generated by the interference of the scanning and stationary beams at a scanning coordinate x is given by

$$I(x) \propto \int O(x')|E_{\text{stat}}(x') + E_{\text{scan}}(x' - x)|^6 dx' \tag{1}$$

Where E_{stat} and E_{scan} is the complex-valued E-field PSF of the stationary and scanning beam and $O(x')$ is the real-valued object function related to the fluorophores distribution in the focal plane. To estimate the wavefront distortions, we need to solve equation (1) for E_{scan}. For simplicity, we denote $a = E_{\text{stat}}(x')$ and $b = E_{\text{scan}}(x' - x)$. The algebraic expansion of $|a + b|^6$, with complex-valued a and b, is

$$\begin{aligned} c = |a + b|^6 = {} & |a|^6 + 9|a|^4|b|^2 + 9|a|^2|b|^4 + |b|^6 \\ & +3(|a|^4 + 3|a|^2|b|^2 + |b|^4) \cdot (ab^* + a^*b) \\ & +3(|a|^2 + |b|^2) \cdot (a^2b^{*2} + a^{*2}b^2) \\ & +a^3b^{*3} + a^{*3}b^3 \end{aligned} \tag{2}$$

The phase difference between the stationary and scanning beam is linearly modulated at a fixed frequency, i.e., $b = b_0 e^{j\omega t}$, where $j = \sqrt{-1}$ and $\omega = 2\pi f$ is the modulation frequency. Then equation (2) becomes

$$\begin{aligned} c(t) = {} & |a|^6 + 9|a|^4|b_0|^2 + 9|a|^2|b_0|^4 + |b_0|^6 \\ & +3(|a|^4 + 3|a|^2|b_0|^2 + |b_0|^4) \cdot \left(ab_0^* e^{-j\omega t} + a^*b_0 e^{j\omega t}\right) \\ & +3(|a|^2 + |b|^2) \cdot \left(a^2b_0^{*2} e^{-j2\omega t} + a^{*2}b_0^2 e^{j2\omega t}\right) \\ & +a^3b_0^{*3} e^{-j3\omega t} + a^{*3}b_0^3 e^{j3\omega t} \end{aligned} \tag{3}$$

Filtering out the components at frequency ω, we have:

$$\{c(t)\}^{(\omega)} = 3(|a|^4 + 3|a|^2|b_0|^2 + |b_0|^4) \cdot \left(ab_0^* e^{-j\omega t} + a^*b_0 e^{j\omega t}\right) \tag{4}$$

Setting the ratio between the two beams so that the stationary beam is much stronger than the scanning beam, i.e., $|a|^2 \gg |b_0|^2$. Then equation (4) can be approximated as:

$$\{c(t)\}^{(\omega)} = 3|a|^4 \cdot \left(ab_0^* e^{-j\omega t} + a^*b_0 e^{j\omega t}\right) \tag{5}$$

Next, we return to equation (1) and the frequency component at ω is:

$$\{I(x,t)\}^{(\omega)} \propto \int 3O(x')|a|^4 \cdot \left(ab_0^* e^{-j\omega t} + a^*b_0 e^{j\omega t}\right) dx' \tag{6}$$

In order to detect the amplitude and phase of the focal field (b_0), the fluorescence signal $I(x,t)$ was fed into a lock-in amplifier working at frequency ω. The two orthogonal outputs X (i.e., in-phase) and Y (i.e., quadrature) of the lock-in amplifier are

$$\begin{aligned} X &= \frac{1}{T}\int_0^T I(t)\cos wt\, dt = \frac{3}{2}\int O(x')|a|^4(ab_0^* + a^*b_0)dx' \\ Y &= \frac{1}{T}\int_0^T I(t)\sin wt\, dt = \frac{3}{2}\int O(x')|a|^4 j(-ab_0^* + a^*b_0)dx' \end{aligned} \tag{7}$$

We can then compute:

$$\begin{aligned} E_{\mathrm{f}}(x) = X + jY &= \int 3O(x')|a|^4 ab_0^* dx' \\ &\propto \int \underbrace{O(x')|E_{\mathrm{stat}}(x')|^4 E_{\mathrm{stat}}(x')}_{\delta-\text{like function}} \cdot E_{\mathrm{scan}}^*(x'-x)dx' \end{aligned} \tag{8}$$

Considering the E-field PSF as an aberrated focus with a strong center and weak side lobes, the 5th order term $O(x')|E_{\mathrm{stat}}(x')|^4 E_{\mathrm{stat}}(x')$ can be regarded as a highly peaked, δ-like function. Also, the measured signal $E_{\mathrm{f}}(x)$ is the convolution of the aberrated focal field E_{scan}^* with a δ-like function. Therefore, $E_{\mathrm{f}}(x)$ can be used to approximate the focal field of the imaging system. Knowing the aberrated E-field PSF, we can use the time reversal symmetry of optical propagation to correct for optical aberrations by phase conjugation. The desired correction pattern for the wavefront-shaping element is the two-dimensional Fourier transform of the measured PSF.

$$\mathcal{F}\{E_{\mathrm{f}}(x)\} = \mathcal{F}\{O(x)|E_{\mathrm{stat}}(x)|^4 E_{\mathrm{stat}}(x)\} \cdot \mathcal{F}\{E_{\mathrm{scan}}(x)\}^* \tag{9}$$

Because the convolution kernel $O(x')|E_{\mathrm{stat}}(x')|^4 E_{\mathrm{stat}}(x')$ in the above analysis is not exactly a δ function, the estimated E-field PSF will approximate but not perfectly match the true focal field. However, after applying the Fourier transform of the estimated E-field PSF to the wavefront shaper, the updated beam will nevertheless be closer to a diffraction-limited spot, which in turn means that the 5th power of its amplitude will resemble a δ function more closely. This allows us to repeat the PSF measurement process and obtain a more accurate PSF in each correction step, thereby iteratively eliminating residual aberrations. We next show that the amplitude of the corrected PSF will be taken to the 5th power after each correction step and finally converge to an ideal focus.

After 1st correction, the focal field will be,

$$E_{\text{1st corr}}(x) = \mathcal{F}^{-1}\left\{\underbrace{\mathcal{F}\{O(x)|E_{\mathrm{stat}}(x)|^4 E_{\mathrm{stat}}(x)\} \cdot \mathcal{F}\{E_{\mathrm{scan}}(x)\}^*}_{\text{1st corr wavefront}} \cdot \underbrace{\mathcal{F}\{E_{\mathrm{PSF}}(x)\}}_{\text{aberration}}\right\} \tag{10}$$

Because the E_{stat} and E_{scan} are equal to the E-field PSF $E_{\mathrm{PSF}}(x)$ within a small lateral range defined by the memory effect and the time reversal property of light propagation implies that $\mathcal{F}\{E_{\mathrm{scan}}\}^* \cdot \mathcal{F}\{E_{\mathrm{PSF}}(x)\} = 1$, we show that corrected PSF in the 1st iteration is equal to:

$$E_{\text{1st corr}}(x) = \mathcal{F}^{-1}\{\mathcal{F}\{O(x)|E_{\text{stat}}(x)|^4 E_{\text{stat}}(x)\}\} = O(x)|E_{\text{stat}}(x)|^4 E_{\text{stat}}(x)$$
$$= O(x)|E_{\text{stat}}(x)|^5 e^{j\varphi_{\text{stat}}(x)} = O(x)|E_{\text{PSF}}(x)|^5 e^{j\varphi_{\text{PSF}}(x)} \quad (11)$$

Next, we repeat this process and scan the corrected scanning beam ($E_{\text{1st corr}}$) against the corrected stationary beam ($E_{\text{1st corr}}$). We can insert $E_{\text{1st corr}}$ into Equation (1) to redefine E_{scan} and E_{stat}. Following exactly the same analysis, we can calculate the 2nd corrective wavefront and add it to the wavefront-shaping device, so the corrected PSF in the 2nd iteration will be:

$$E_{\text{2nd corr}}(x) = \mathcal{F}^{-1}\left\{\underbrace{\mathcal{F}\{O(x)|E_{\text{1st corr}}(x)|^4 E_{\text{1st corr}}(x)\} \cdot \mathcal{F}\{E_{\text{1st corr}}(x)\}^*}_{\text{2nd corr wavefront}}\right.$$
$$\left. \cdot \underbrace{\mathcal{F}\{E_{\text{1st corr}}(x)\}}_{\text{residual wavefront}}\right\} = O(x)|E_{\text{1st corr}}(x)|^5 e^{j\varphi_{\text{1st corr}}(x)} \quad (12)$$
$$= O(x)(O(x)|E_{\text{PSF}}(x)|^5)^5 e^{j\varphi_{\text{1st corr}}(x)}$$

We can see that by iteratively sensing the focal field and correcting the excitation beam, the PSF converges towards its 5th power of the previous correction. Applying this nonlinear factor to any aberrated focus will help any dominant mode to prevail against other weaker side lobes and finally result in a near-diffraction-limited PSF.

It is clear from this analysis that the number of correction steps required to converge on the diffraction-limited PSF will depend strongly on the original shape of the scattered PSF. The presence of a dominant mode will allow faster convergence, compared to the case where multiple modes have comparable intensities. Moreover, as can be seen from equation (11-12), if the imaging sample is sparse and contains point-like fluorescent structures (i.e. $O(x)$ is a δ-like function), the convergence rate will be faster than in the uniform fluorescence case (i.e. $O(x)$ is a uniform function).

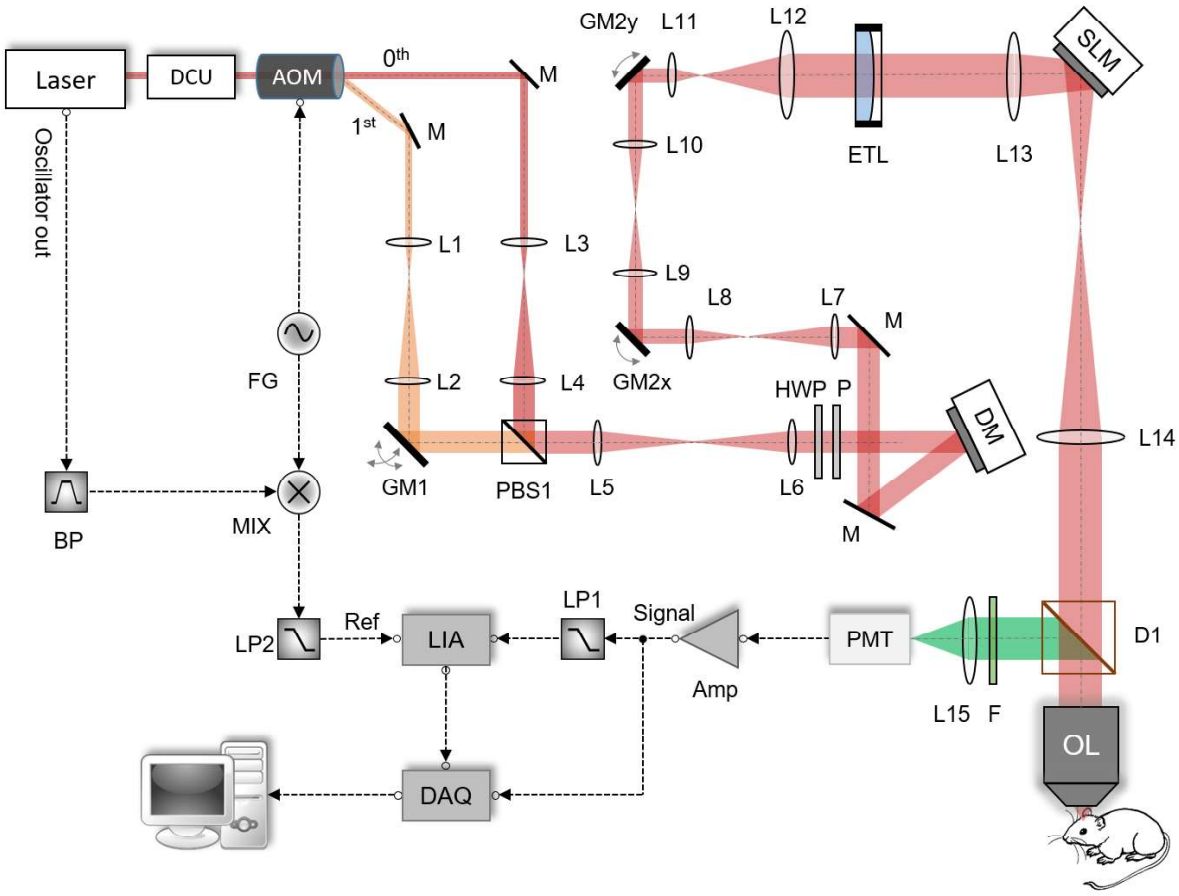


Figure S1. Detailed schematic diagram of the conjugate AO-3P microscope. Laser: a two-stage optical parametric amplifier (Orpheus-F, Light Conversion) pumped by a 40 W femtosecond laser (Carbide, Light Conversion); DCU: dispersion compensation unit; AOM: acoustic optical modulator (AOMO 3080-122, Crystal Technology) driven by a RF signal generator (AODS20160-8, Crystal Technology); L1-L15: lenses; M: mirror; GM1: galvo scanning mirrors (GVS002, Thorlabs); PBS1: polarizing beam splitter; HWP: half wave plate; P: polarizer; DM: deformable mirror (DM97-15, Alpao); 4*f* system-1: L7 (f = 230 mm, consisted of LA1484-C and LA1464-C, Thorlabs) and L8 (f=50 mm, AC127-050-C, Thorlabs); GM2x and GM2y: X and Y galvo scanner (6215H, Cambridge Technology); 4*f* system-2: L9 and L10 (f=100 mm, #45-806, Edmund Optics); 4*f* system-3: L11 (f=50 mm, SL50-2P2, Thorlabs) and L12 (f=250 mm, AC508-250-C, Thorlabs); ETL: electrical tunable lens (EL-16-40-TC-NIR-5D-C, Optotune); 4*f* system-4: L13 and L14 (L13: f=200 mm, #47-271; L14: f=150 mm, #47-380, Edmund Optics); OL: objective lens (NA 1.05, 25×, XLPLN25XWMP2, Olympus); SLM: spatial light modulator (HSP1920-1064-HSP8, Meadowlark Optics); 4*f* system-5: L14 and OL; D1: dichroic mirror (FF875-Di01-25x36, Semrock); F: band pass filters (FF02-447/60-25, FF03-525/50-25 or FF01-593/46-25, Semrock) and short pass filter (FF01-770/SP-25, Semrock); PMT: photomultiplier tube (H7422, Hamamatsu); Amp: current amplifier (14 MHz bandwith, $\times 10^4$ gain, DHCPA-200, Femto); LP1: a low-pass and band-pass filter (BLP-1.9+ and ZFBP-400K-S+, Mini-Circuit); LIA: lock-in

amplifier (SR844, Stanford Research Systems); FG: function generator (DS345, Stanford Research Systems); BP: band pass filter (BBP-60+, Mini-Circuit); MIX: frequency mixer (homemade); LP2: low pass filter (BLP-1.9+, Mini-Circuit); DAQ: data acquisition device (vDAQ, Vidrio Technologies).

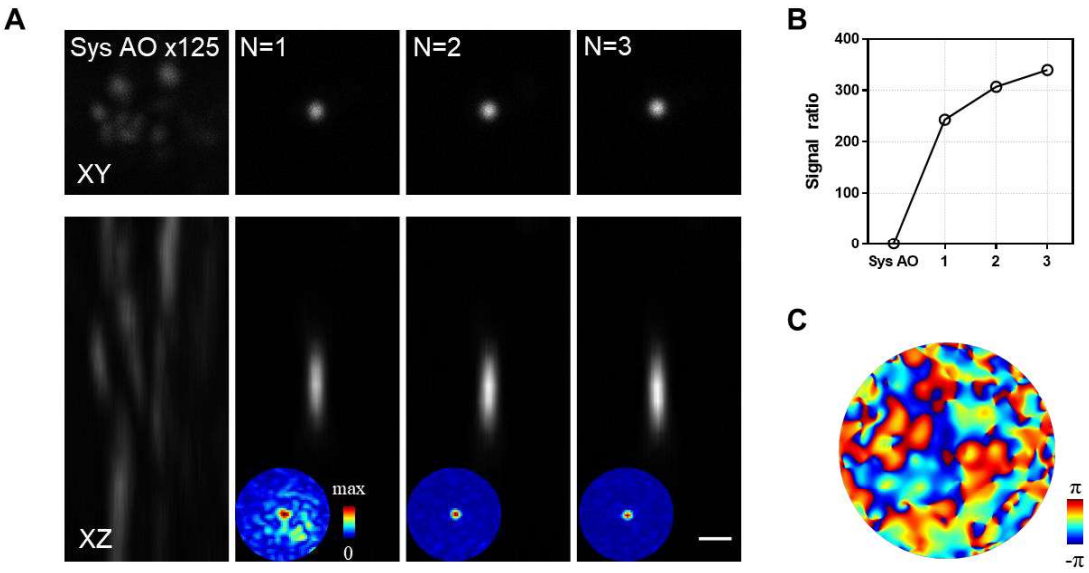


Figure S2. *In vitro* evaluation of iteration effect on fluorescence signal improvement. (A) Lateral and axial views of a 1-μm-diameter fluorescent bead at 300 μm below an excised intact skull (100μm-thickness) imaged with only system aberration corrected and with full correction running a total N (N=1-3) iterations. The insets display the measured PSF magnitude before applying the Nth correction. Scale bar: 2μm. (B) Fluorescence signal improvement (full/system) with different iterations. (C) Corrective phase pattern finally measured.

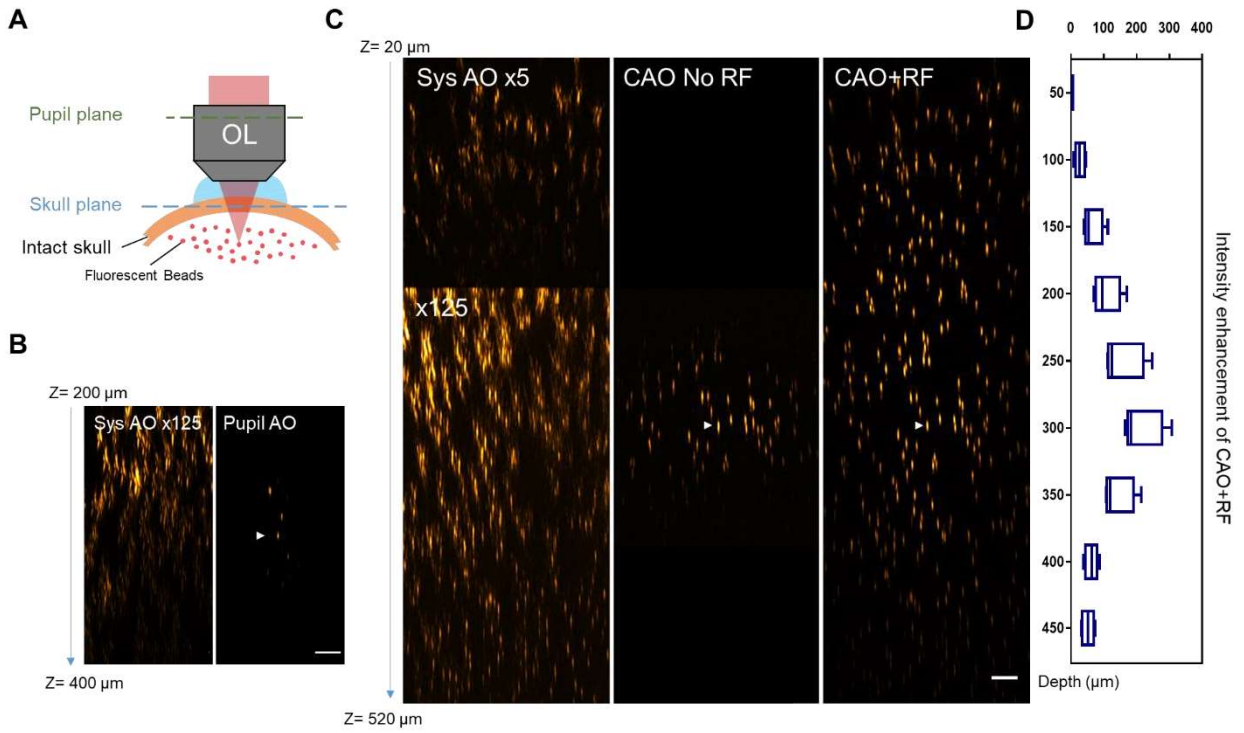


Figure S3. *In vitro* validation of conjugate AO with remote focusing. (A) Schematic diagrams illustrating the *in vitro* preparation. Red fluorescent beads of 1µm diameter were imaged through an excised intact mouse skull of 100µm thickness. SLM is either conjugated to the pupil plane of objective lens (Pupil AO) or the middle of intact skull (Conjugate AO). (B) XZ views of fluorescent beads (200-400 µm below pia) with system correction only (left) and with full correction of pupil AO (right). The arrow head indicates the fluorescent bead used for aberration measurement. Scale bar: 20 µm. (C) XZ views of fluorescent beads (20-520 µm below pia) with system correction (Sys AO) only (left), full correction of conjugate AO (CAO) without remote focusing (RF) (middle) and with RF (right). Single CAO correction is based on the measurement from the fluorescent bead indicated by the arrow head. Scale bar: 20 µm. (D) Signal intensity enhancement of fluorescent beads at different depths for "CAO+RF" image of (C) compared with that in "Sys AO" image. N = 5 independent measurements. Data are presented as Box and whiskers with 10-90 percentile whisker and median as the center.

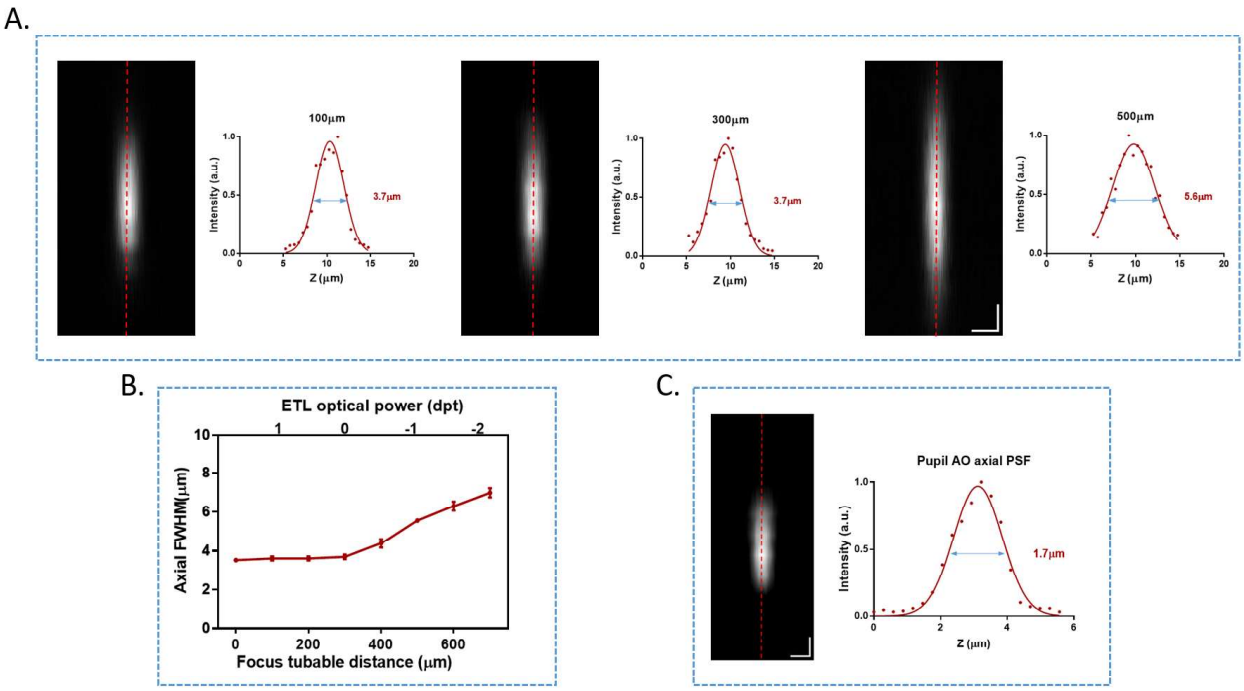


Figure S4. Experimental measurements of axial resolutions for the conjugate α-FSS-3PM (NA: 0.70) and pupil α-FSS-3PM (NA: 1.05). (A) Representative XZ views of single fluorescent bead (0.2 μm-diameter) in agarose and line-profile fittings of remote focusing at 100, 300, 500 μm focal point tuning distance, respectively. Scale bar: 1 μm. (B) Axial resolution of conjugate α-FSS-3PM over focal point tuning distance from 0 to 700 μm. Note that the axial resolution remained about 3.7 μm over the first 400 μm tuning distance of focal point, close to the theoretical resolution ~3.5 μm, and increased gradually up to 6.9 μm at 700 μm. The degradation of resolution was due to that the laser beam became larger than SLM when the focal point was tuned beyond 400 μm, causing the reduction of effective NA and enlargement of PSF. In future work, a SLM of larger aperture will be used to extend the near diffraction limited resolution over greater tuning range of focal point. (C) Axial resolution of pupil FDSS-AO-3PM at the natural focal plane of objective lens. Scale bar: 0.5 μm.

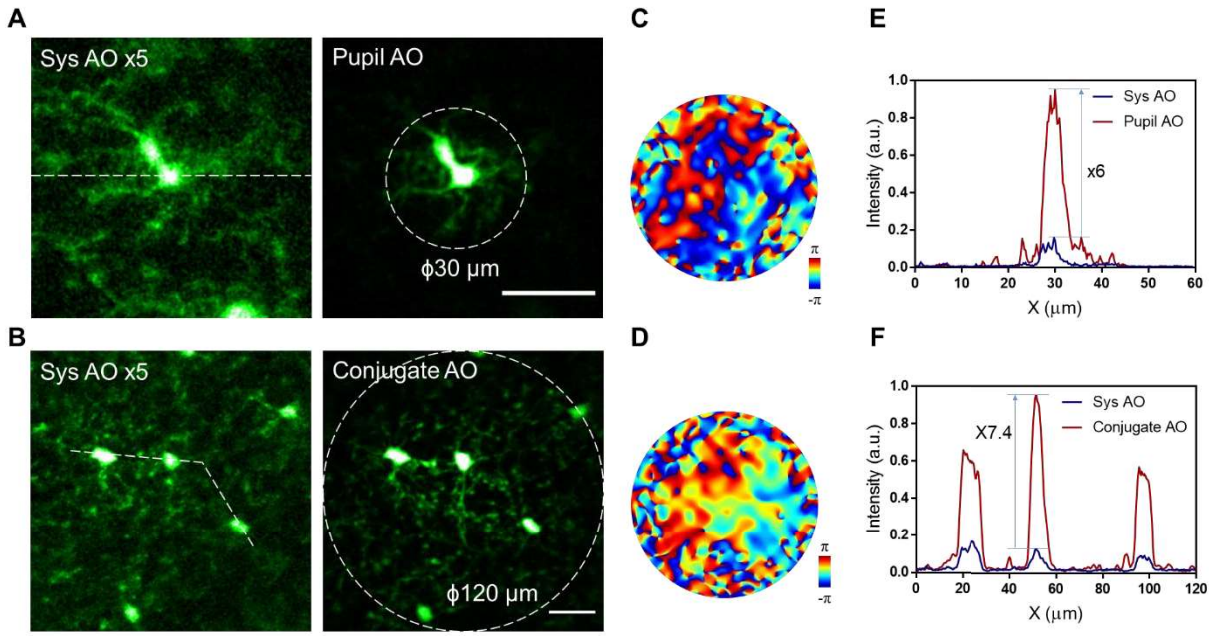


Figure S5. ***In vivo*** **imaging of microglia through intact skull with pupil and conjugate AO.** (A-B) MIP of stack images (Z=280-320μm) of microglia in Cx3Cr1-GFP mouse (2-month-old) through intact skull (100μm thick) with system correction only (left) and full correction (right). Full correction in (A) and (B) are based on pupil AO and conjugate AO, respectively. The dashed circles indicate the effective FOV after full AO correction. Scale bar: 20μm. (C-D) Correction phase pattern of SLM used for full correction in (A) and (B), respectively. (E-F) Signal profiles along the dashed lines in (A) and (B), respectively.

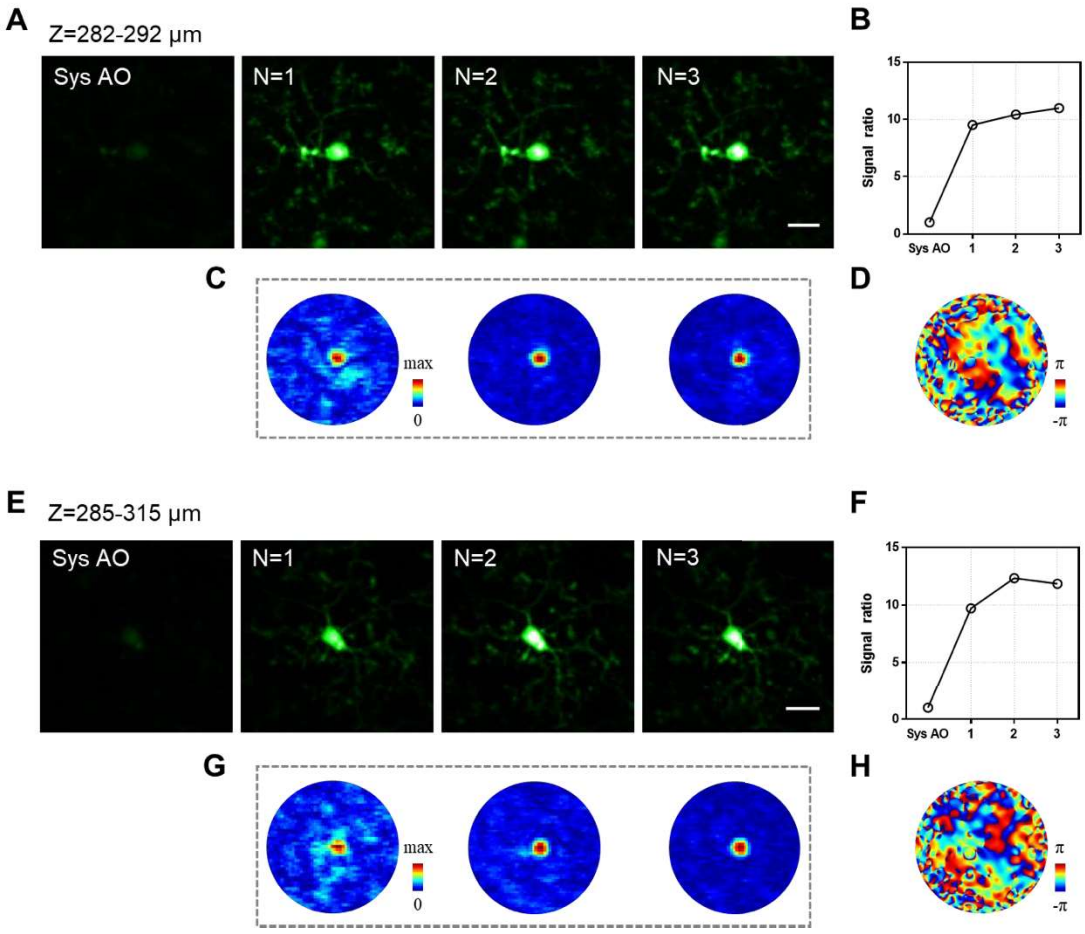


Figure S6. *In vivo* evaluation of iteration effect on fluorescence signal improvement. (A) and (E) Lateral and axial views of GFP-labeled microglia imaged through the intact skull (110μm thickness) with only system aberration corrected and with full correction running a total N (N=1-3) iterations. Scale bar: 10μm. (B) and (F) Fluorescence signal improvement (full/system) with different iterations. (C) and (G) Measured PSF magnitude before applying the Nth correction. (D) and (H) Corrective phase pattern finally measured. Note that in panel (F), the signal enhancement ratio decreased slightly from 12.3× to 11.9× after the 3rd iteration, which is negligible and could be attributed to the fluctuation of laser power.

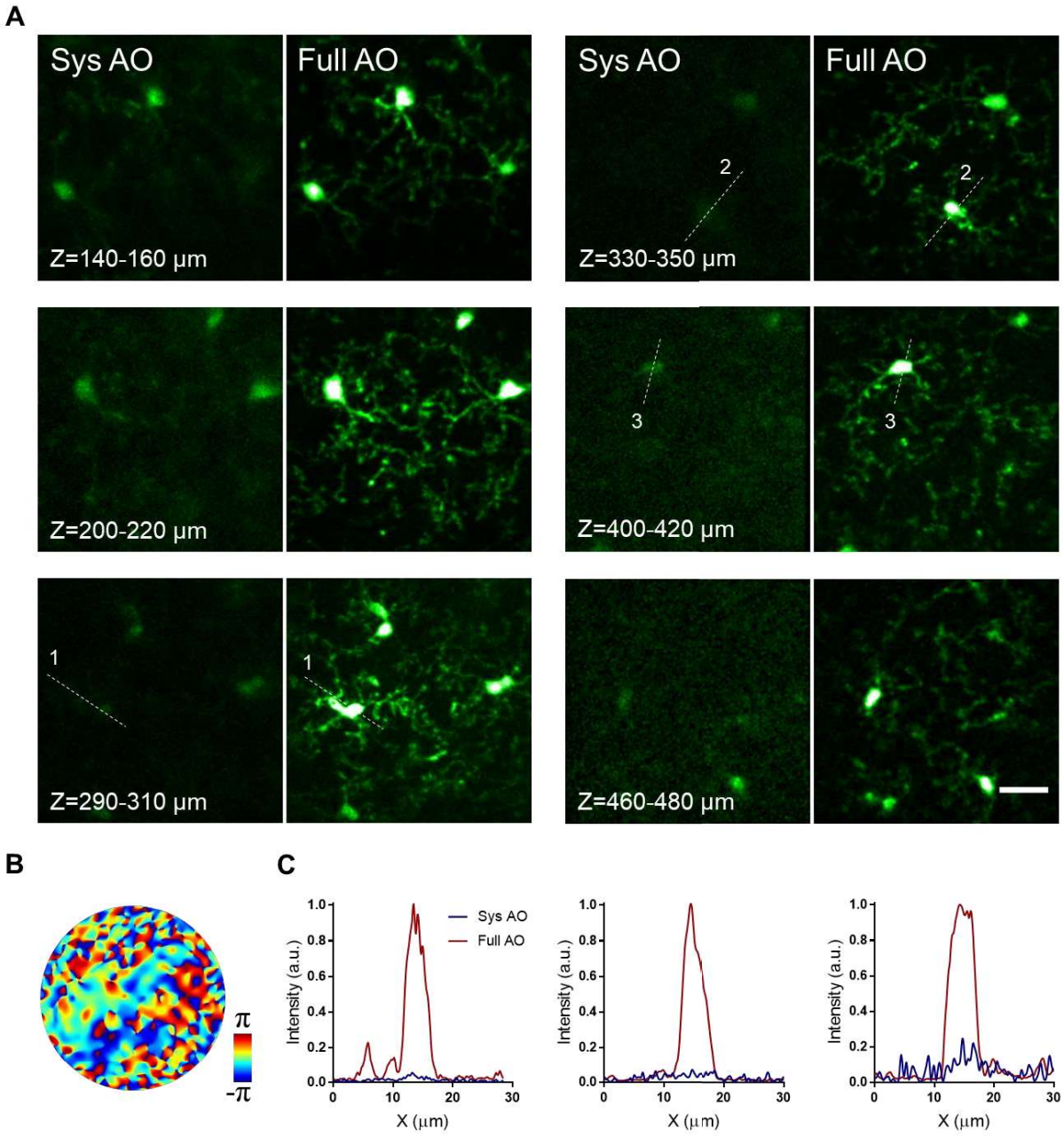


Figure S7. Single correction of conjugate AO recovers subcellular resolution over a large volume *in vivo*. (A) *In vivo* imaging of microglia in Cx3Cr1-GFP mouse (2-month-old) through intact skull (100μm thick) with system correction only and with full correction of conjugate AO. Scale bar: 20 μm. (B) Phase pattern of SLM for full AO correction in (A) is based on focus sensing at 300 μm below pia. (C-E) Signal profiles along the dashed lines in (A).

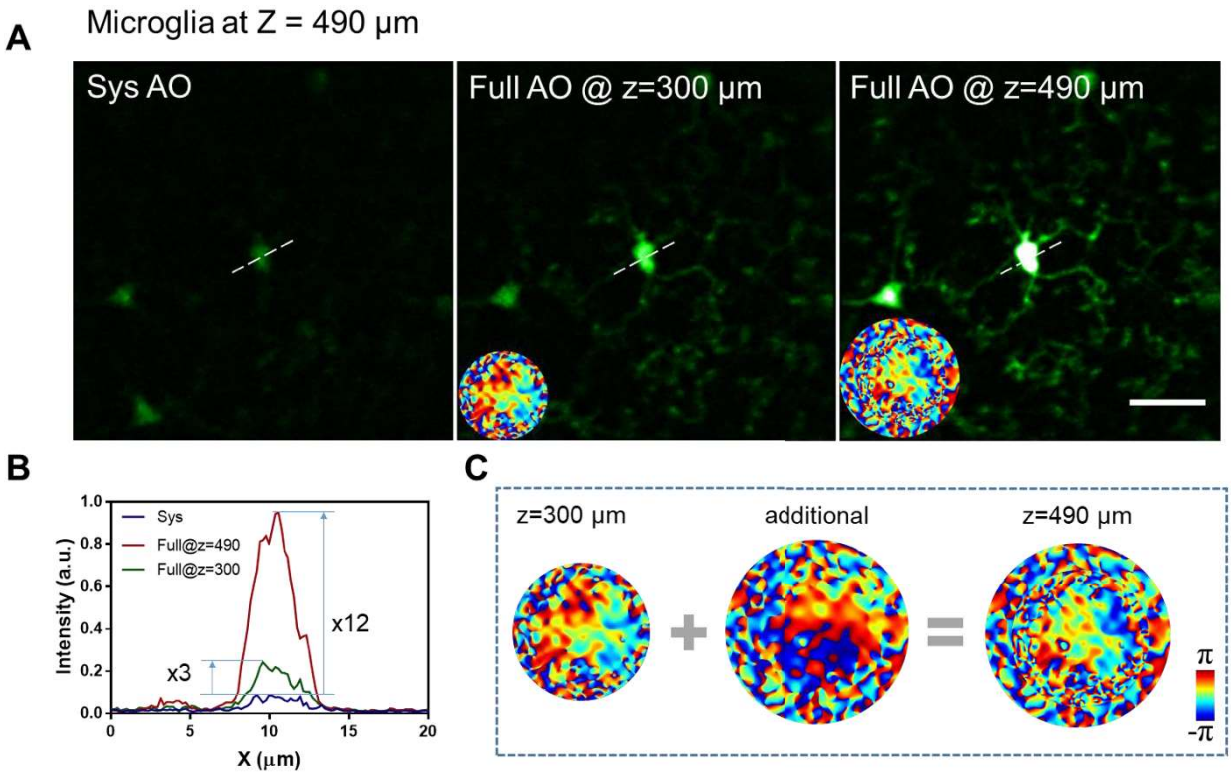


Figure S8. Validation of pre-compensation AO strategy for improving *in vivo* imaging in deeper cortical regions. (A) *In vivo* imaging of microglia at 490μm below pia of Cx3Cr1-GFP mouse through intact skull with system correction only (left) and with full correction of conjugate AO (middle and right). Insets in the middle and right images are the corresponding correction phase patterns of SLM, which were based on α-FSS at Z=300 μm and Z=490 μm, respectively. The images are MIP of stack images at Z=480-500 μm. Scale bar: 20 μm. (B) Signal profiles along the dashed lines in (A). (C) Focus sensing at Z=490 μm was conducted with pre-compensation of the Z=300μm phase pattern, generating the additional phase pattern. The final correction phase pattern for Z=490μm image is derived by summing up the Z=300μm phase with the additional phase.

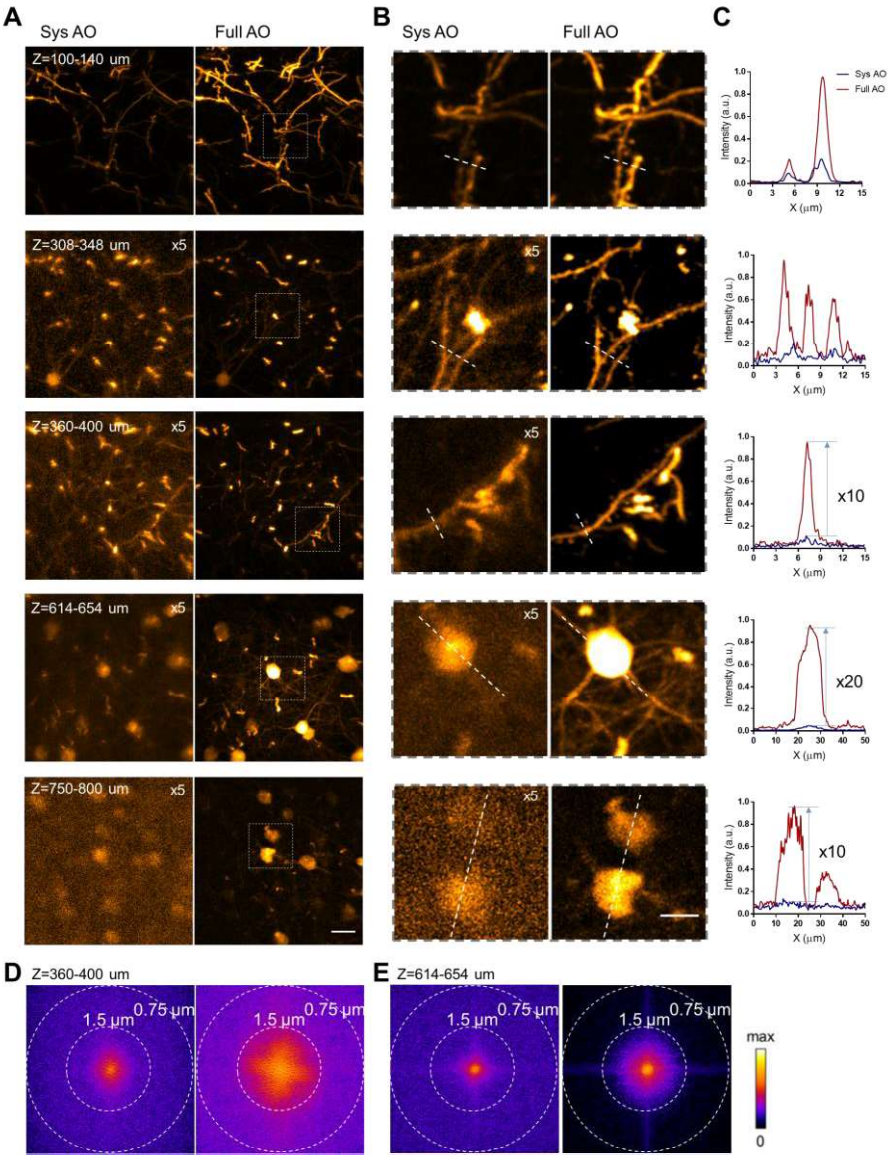


Figure S9. Conjugate α-FSS-3PM improves *in vivo* imaging of neuronal structures in mouse cortices through intact skull. (A) MIP of stack images of YFP-labeled neurons at different depths with system correction only (left) and with full AO correction (right). Image of system AO was digitally enhanced as indicated for better visualization. Scale bar: 20 μm. (B) Magnified views of neuronal structures in box regions as indicated in (A). Scale bar: 10 μm. (C) Signal profiles along the dashed lines in (B). Improvement of signal intensity is given. Note that the gain of signal increases with the imaging depth. With increase of imaging depth, the excitation beam covers larger area of skull, which contributes for the larger aberration/ scattering and causes rapid drop of signal intensity and resolution. After α-FSS AO correction, both signal intensity and resolution are increased significantly, and therefore higher gain of signal were obtained at a deeper depth. (D-E) Spectral power in spatial frequency space for the images in (A) at depth of Z=360-400μm and Z=614-654μm, respectively. Left: system AO; right: full AO.

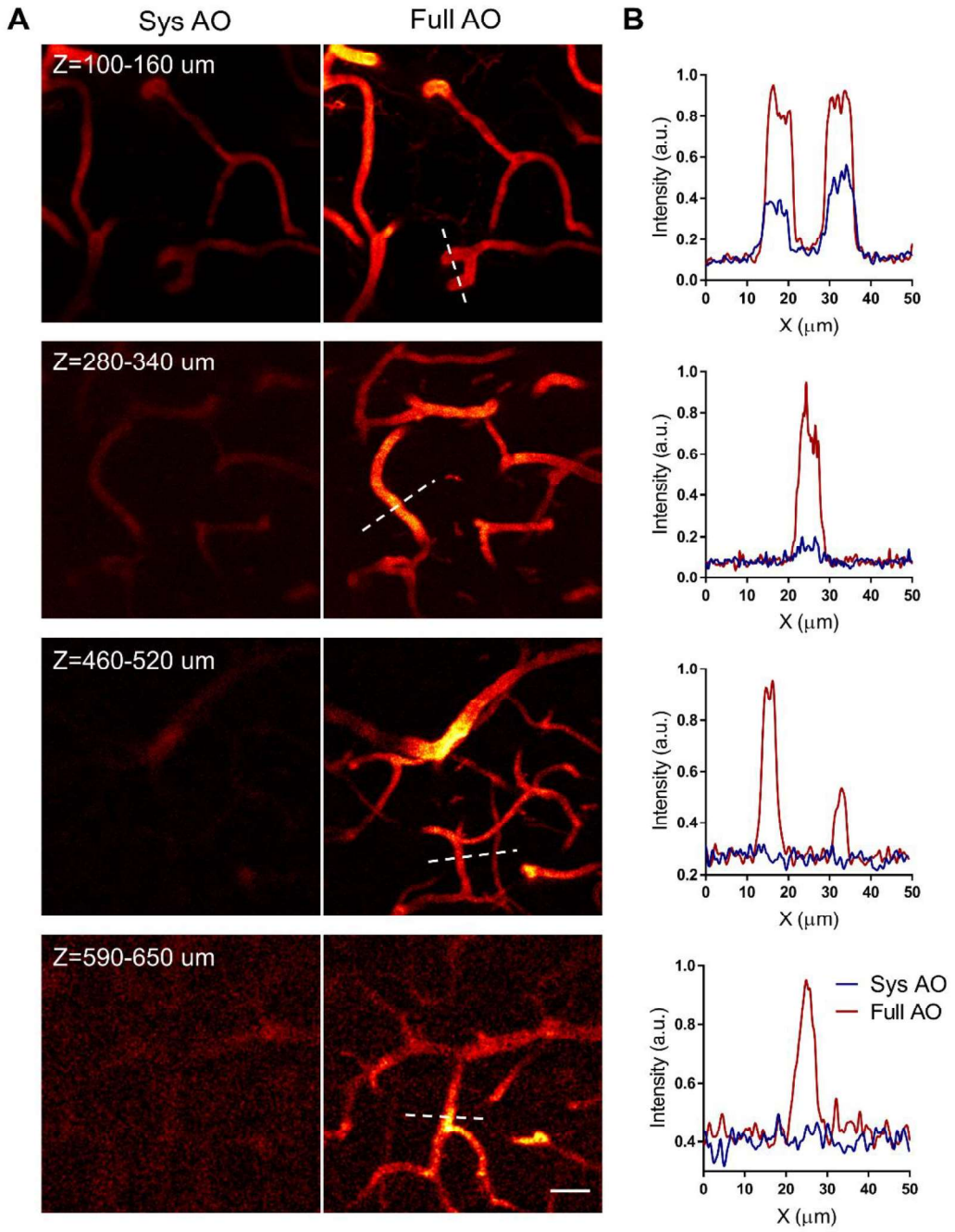


Figure S10. Conjugate α-FSS-3PM improves *in vivo* imaging of blood vessels in mouse cortices through intact skull. (A) MIP of stack images of Texas-Red-labeled vessels at different depths with system correction only (left) and with full AO correction (right). Scale bar: 20 μm. (B) Signal profiles along the dashed lines in (A).

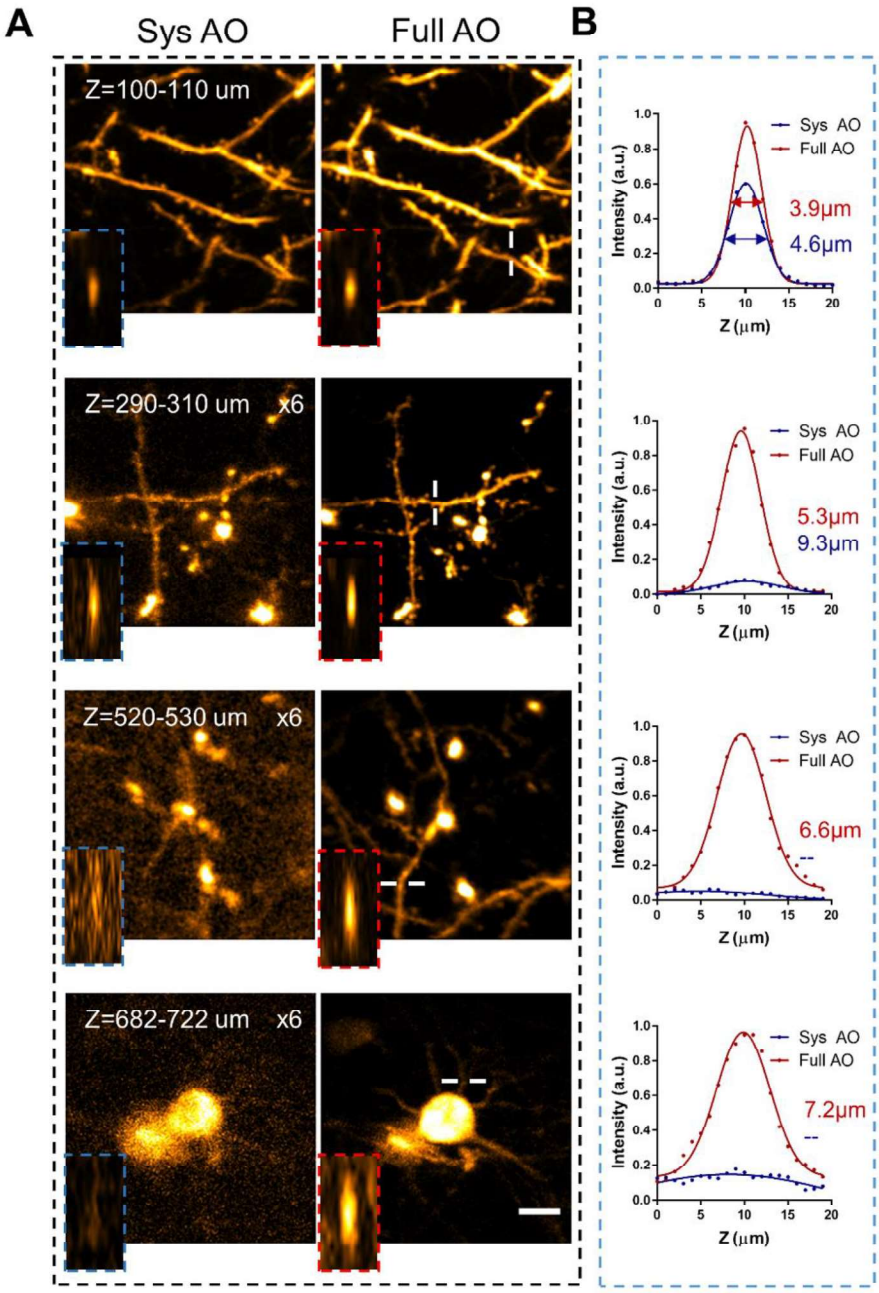


Figure S11. Conjugate α-FSS-3PM restores near-diffraction-limited resolution at depths for *in vivo* imaging of mouse cortices through intact skull. (A) MIP of stack images of YFP-labeled neurons at different depths with system correction only (left) and with full AO correction (right). Insets show the XZ view of dendrites through the dashed lines. Scale bar: 10 μm. (B) Axial resolutions with system correction (blue) and full AO correction (red) measured from the signal profiles of the XZ views in (A). Note that the resolution after AO correction became poorer with increasing depth. This could be caused by the decrease of effective NA due to (1) lost of marginal rays from the scattering of brain volume and (2) beam clipping at the SLM using remote focusing when imaging deep region.

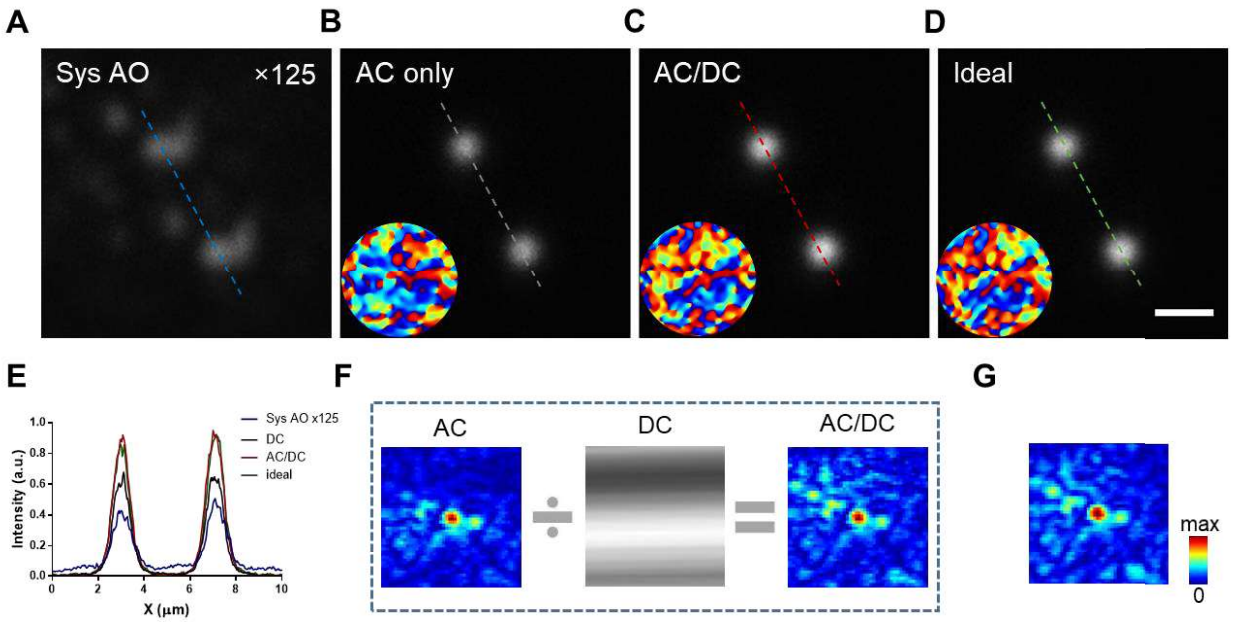


Figure S12. Validation of α-FSS with fluctuating fluorescence intensity. (A-D) Imaging of fluorescent beads at 300μm below an intact mouse skull with system correction only (A) and with full AO correction (B-D). During focus sensing in (B) and (C), the excitation laser power was continuously adjusted to induce fluctuation in fluorescence intensity mimicking the scenario of calcium imaging. The insets in (B-C) are the corresponding corrective wavefronts, which are calculated by using the raw (AC) magnitude of PSF for (B), DC-normalized (AC/DC) magnitude of PSF for (C). During focus sensing in (D), the excitation power remains unchanged and the corrective wavefront (inset) is calculated by using the raw magnitude of PSF and serves as an ideal case for full AO correction. Scale bar: 2 μm. (E) Signal profiles along the dashed lines in (A-D). (F) Correction of PSF magnitude is achieved by normalizing its raw (AC) magnitude to the low pass filtered fluctuating fluorescence intensity (DC) during direct focus sensing (details in *Methods and Materials*). (G) PSF magnitude of ideal case (D), which resembles the corrected PSF magnitude in (F).

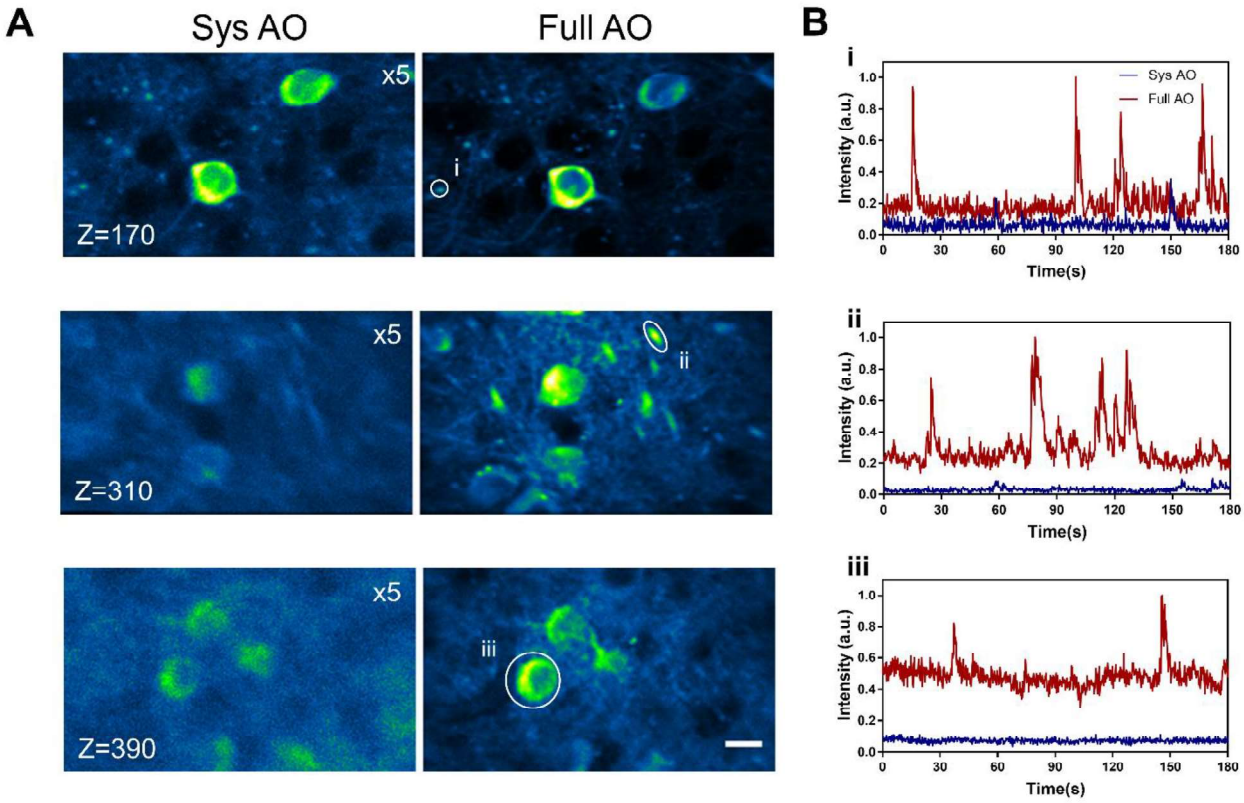


Figure S13. AO-assisted *in vivo* multi-plane calcium imaging of neuronal activity through intact skull. (A) Multi-plane calcium imaging of spontaneous neuronal activity in the somatosensory cortex of a CCK-GCaMP6s mouse (4-month-old) with system correction only (left) and with full correction (right). Three planes at different depths below pia were sequentially captured at 10.7Hz frame rate. Images are shown as the average-intensity projection of 900 frames. The image of system AO was digitally enhanced 5 fold as indicated to provide better visualization. Scale bar: 10 μm. (B) Fluorescence traces extracted from the ROIs of different neuronal compartments (i and ii: dendrites, iii: soma) as indicated in (A).

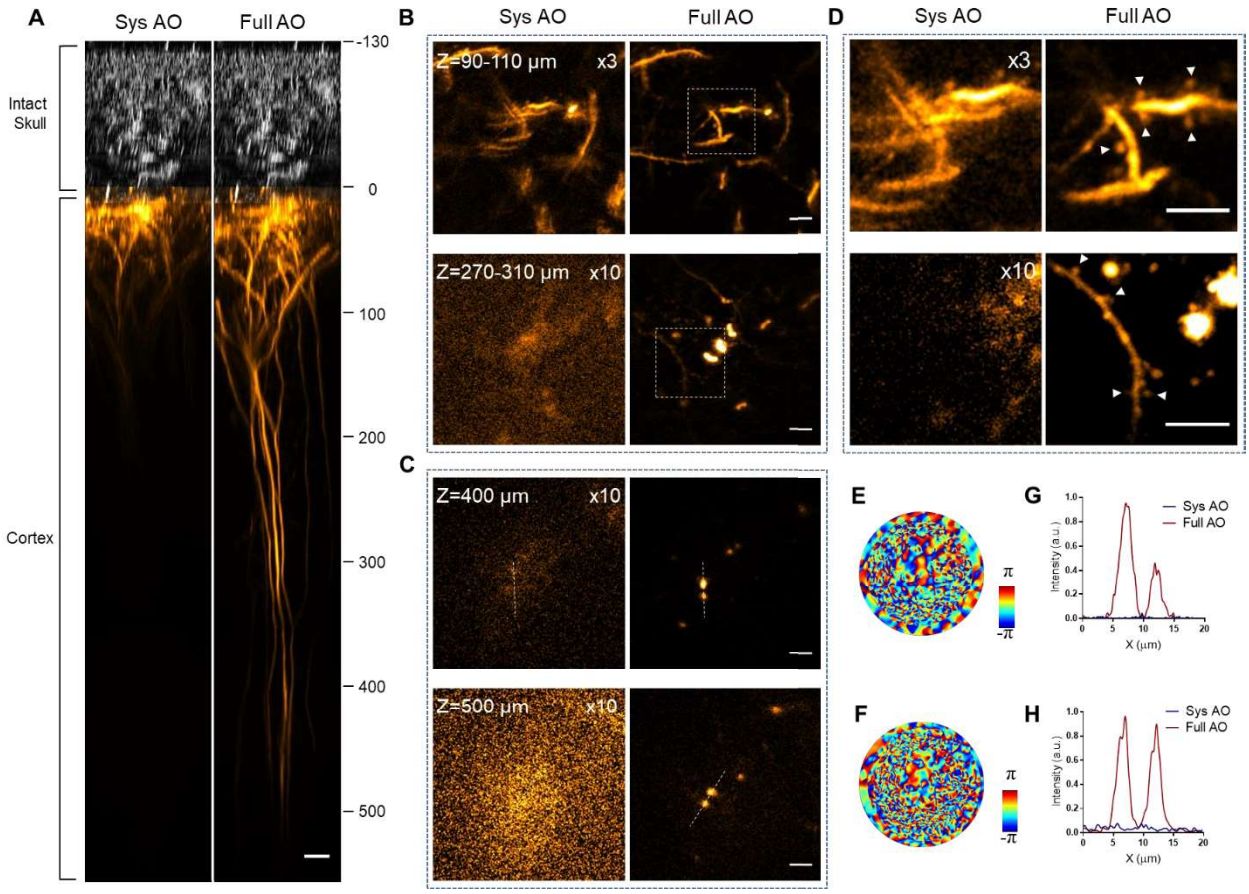


Figure S14. Conjugate α-FSS-3PM recovers subcellular resolution for *in vivo* cortical imaging through the intact skull of aged mice. (A) XZ MIP images of the pyramidal neurons in Thy1-YFP mouse (12-month-old) through a 140μm-thick intact skull with system correction only (left) and full correction (right). Scale bar: 20μm. (B-C) XY MIP of the stack images at different depths below pia in (A). Scale bar: 20 μm. (D) Magnified views of neuronal dendrites from box regions in (B). White arrow heads indicate the dendritic spines. Scale bar: 10μm. (E-F) Correction phase patterns of SLM for the full AO image in (C). (G-H) Signal profiles along the dashed lines in (C). Note that due to the severe scattering and aberration induced by the irregular skull in aged mice, the conventional 3P imaging (system AO) is unable to produce reasonable resolution at depths over 100 μm below pia as shown in (A). With α-FSS to correct for the aberrations, however, the imaging resolution was recovered to subcellular level and the fluorescence signal was substantially improved, enabling dendritic spines to be visualized clearly. The complex pattern of corrective phase indicates that α-FSS compensates the higher-order aberrations to achieve high-resolution imaging through the thicker intact skull of aged mice. By using conjugate AO, we successfully maintained high-resolution 3P imaging over 500 μm below pia. It should be noted that the corrective FOV became smaller for *in vivo* imaging through the 140μm-thick intact skull than the 100μm skull because the single-layer conjugated AO approach works at its best for a concentrated thin layer of aberrated medium (Park, J.-H., Sun, W. & Cui, M. High-resolution in vivo imaging of mouse brain through the intact skull. *Proc. Natl. Acad. Sci.* **112**, 9236–9241 (2015)).

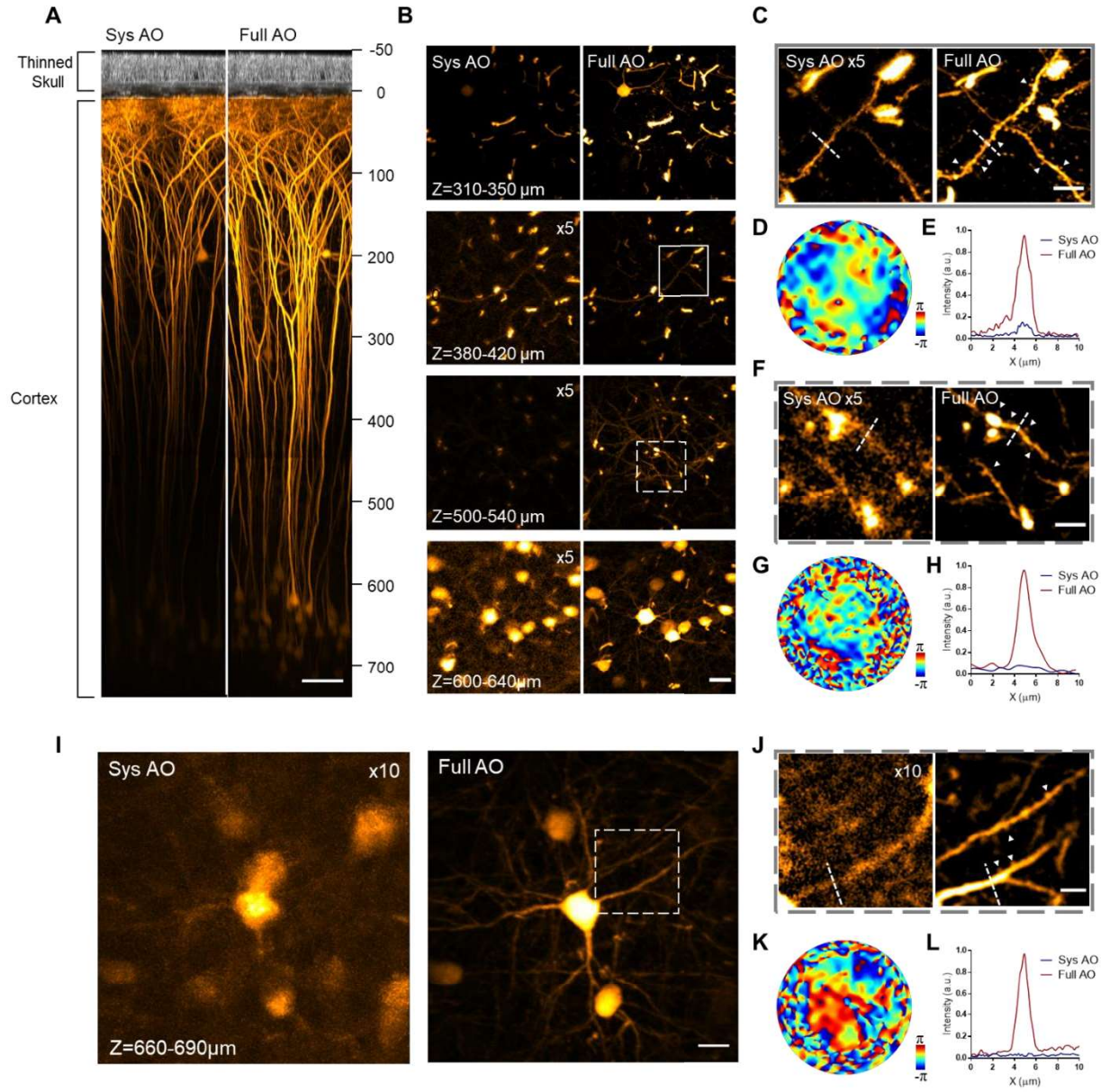


Figure S15. Conjugate α-FSS-3PM improves *in vivo* cortical imaging through a thinned skull window. (A) XZ MIP images of the pyramidal neurons in Thy1-YFP mouse (2-month-old) through a thinned skull window (50 μm thickness) with system correction only (left) and full correction (right). The zero depth is set just beneath the pia. Scale bar: 50 μm. (B) XY MIP of the stack images at different depths below pia in (A). Scale bar: 20 μm. (C) and (F) Magnified views of neuronal dendrites from box region in (B). White arrow heads indicate the dendritic spines. Scale bar: 10μm. (D) and (G) The corresponding correction phase pattern of SLM for the full AO image in (C) and (F), respectively. (E) and (H) Signal profiles along the dashed lines in (C) and (F), respectively. (I) XY MIP of the stack images at 660-690μm below pia. Scale bar: 10 μm. (J) Magnified views of neuronal dendrites from box region in (I). Scale bar: 5 μm. (K) Correction phase pattern of SLM for the full AO image in (I). (L) Signal profiles along the dashed lines in (J).

Note that despite conventional 3P imaging achieving a larger penetration depth through thinned skull than through intact skull, its spatial resolution still degrades rapidly at depth, being insufficient to identify tiny neuronal structures such as dendritic spines beyond 400 μm below the pia as shown in (A). We further demonstrated that the conjugate α-FSS AO approach also significantly improved *in vivo* cortical imaging through a thinned skull window, recovering high imaging resolution over a large depth (>700 μm) similar to that of intact skull (Fig. S15 and Fig. S16). Here, the FOV improvement for 50-μm thinned skull was not significant in comparison with 100 μm intact skull. It could be attributed to the imperfection of skull thinning procedure that caused uneven thickness across the thinned skull and introduced additional optical inhomogeneity. Moreover, functional calcium imaging through a thinned skull window is also drastically improved after full AO correction (Fig. S17), and we achieved highly-sensitive and accurate recording of spontaneous calcium activity from neuronal somata and dendrites on different planes up to 600μm below the cortical surface (Fig. S18).

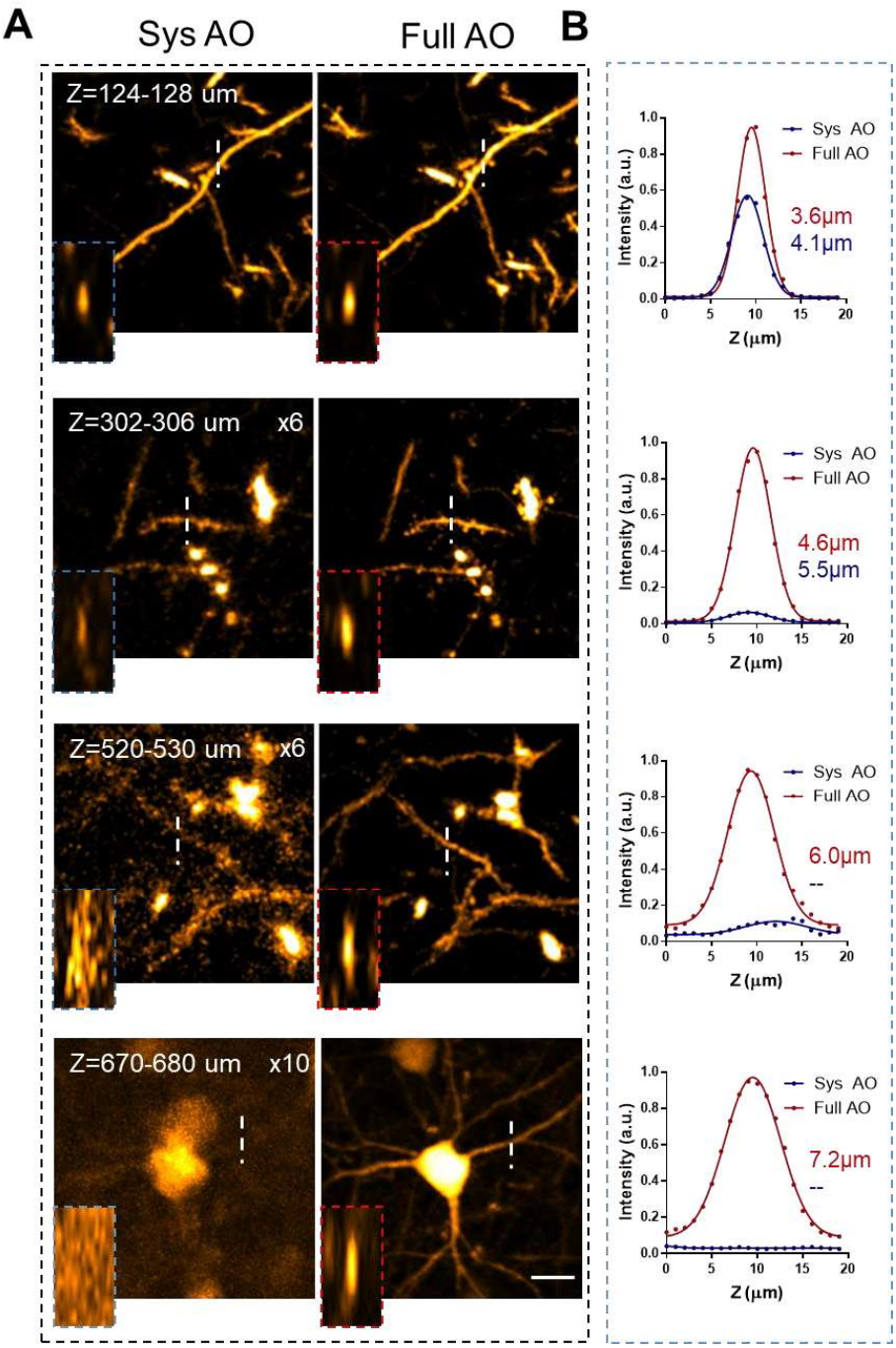


Figure S16. Conjugate α-FSS-3PM restores near-diffraction-limited resolution at depths for *in vivo* imaging of mouse cortices through thinned skull. (A) MIP of stack images of YFP-labeled neurons at different depths with system correction only (left) and with full AO correction (right). Insets show the XZ view of dendrites through the dashed lines. Scale bar: 10 μm (B) Axial resolutions with system correction (blue) and full AO correction (red) measured from the signal profiles of the XZ views in (A).

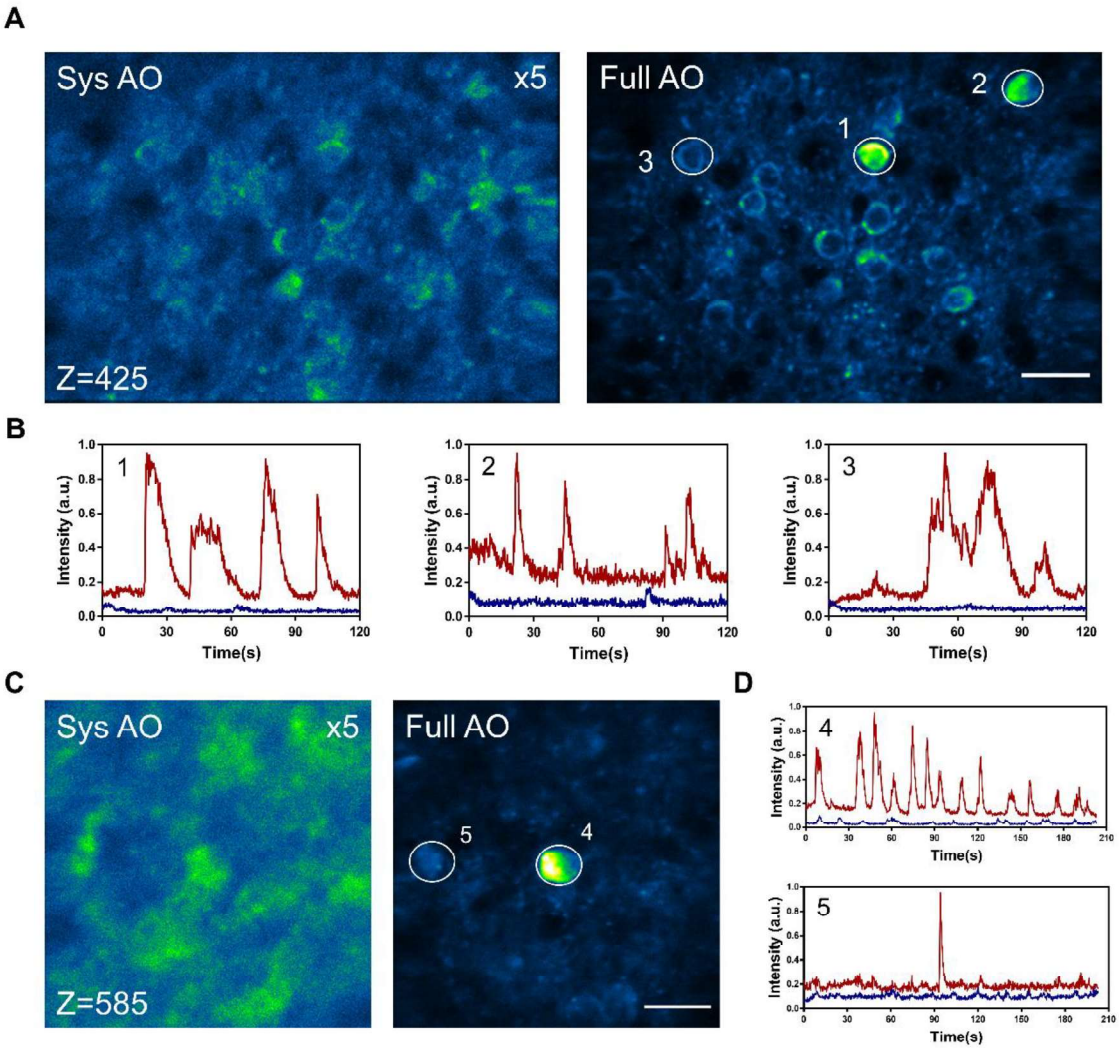


Figure S17. Conjugate α-FSS-3PM improves *in vivo* functional calcium imaging of neuronal activity through a thinned skull window. (A) *In vivo* 3P calcium imaging of spontaneous neuronal activity at 425μm below the pia of the somatosensory cortex of a CCK-GCaMP6s mouse (4-month-old) through thinned skull (50 μm thickness) with system correction only (left) and full AO correction (right). Images are acquired at 5fps and shown as an average-intensity projection of 600 frames. Scale bar: 20 μm. (B) Fluorescence traces extracted from the numbered neuron bodies as shown in (A). (C) *In vivo* calcium images at 585μm below pia of somatosensory cortex acquired at 4.43fps and shown as average-intensity projections of 900 frames. (D) Fluorescence traces extracted from the numbered neuron bodies as shown in (C). Scale bar: 20 μm.

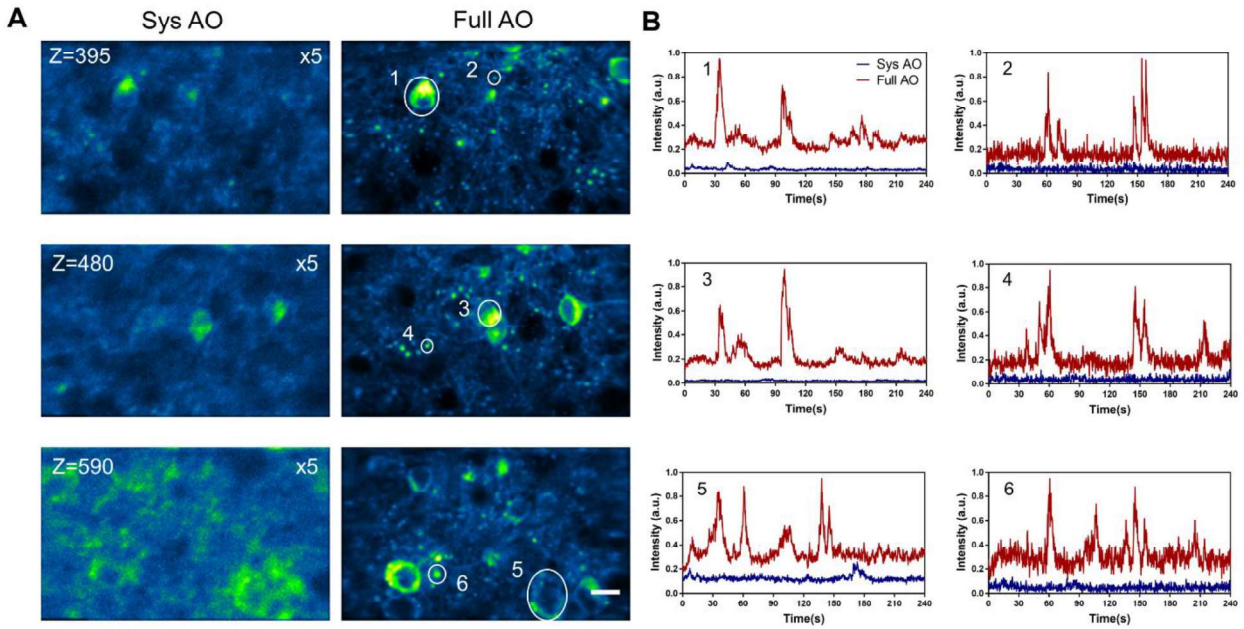


Figure S18. AO-assisted *in vivo* multiplane calcium imaging of neuronal activity through a thinned skull window. (A) Multiplane calcium imaging of spontaneous neuronal activity in the somatosensory cortex of a CCK-GCaMP6s mouse (4-month-old) through a thinned skull window (50 μm thickness) with system correction only (left) and with full correction (right). Three planes at different depths below pia were sequentially captured at 10.7Hz frame rate. Images are shown as average-intensity projections of 900 frames. Scale bar: 10 μm. (B) Fluorescence traces extracted from the ROIs of different neuronal compartments as indicated in (A).

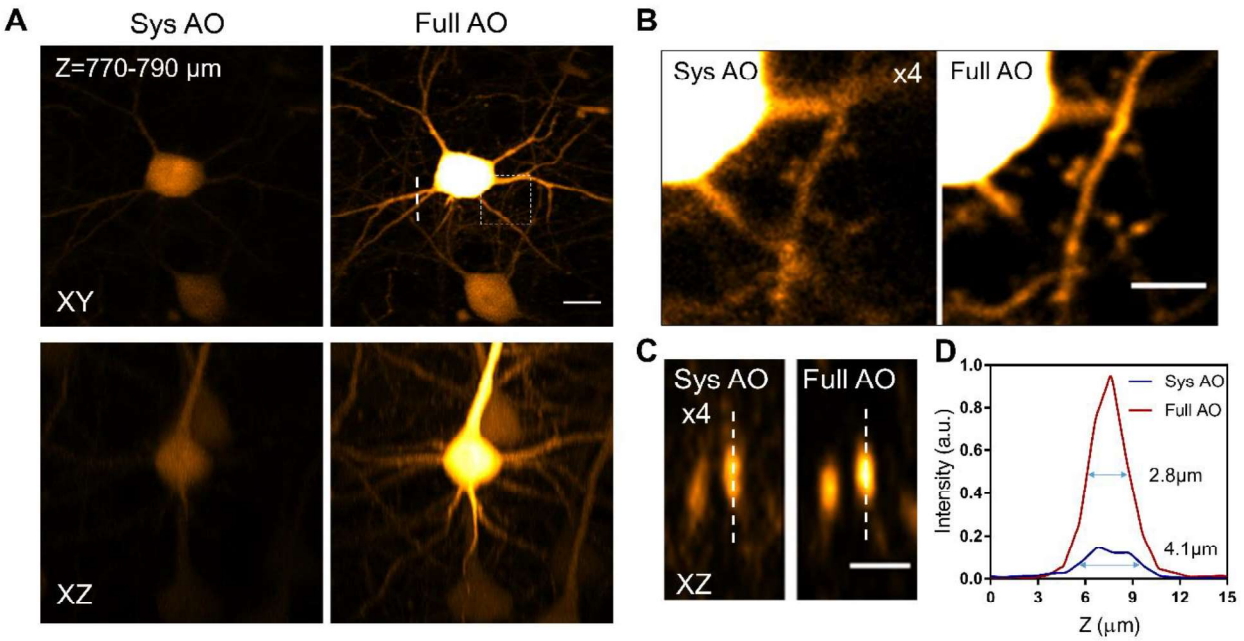


Figure S19. Pupil α-FSS-3PM improves *in vivo* imaging of neuronal structures in deep cortical layers through an open skull window. (A) MIP of stacked images of pyramidal neurons in deep cortical layers (770-790 μm) with system correction only (left) and with full correction (right). Images are shown as XY (upper) and XZ views. Scale bar: 10μm. (B) Magnified views of dendrites from the boxed region indicated in (A). Scale bar: 5 μm. (C) XZ view of dendrites through the plane defined by the dashed line in (A). Scale bar: 2 μm. (D) Signal profiles along the dashed lines in (C), which indicates the axial resolutions in terms of FWHM.

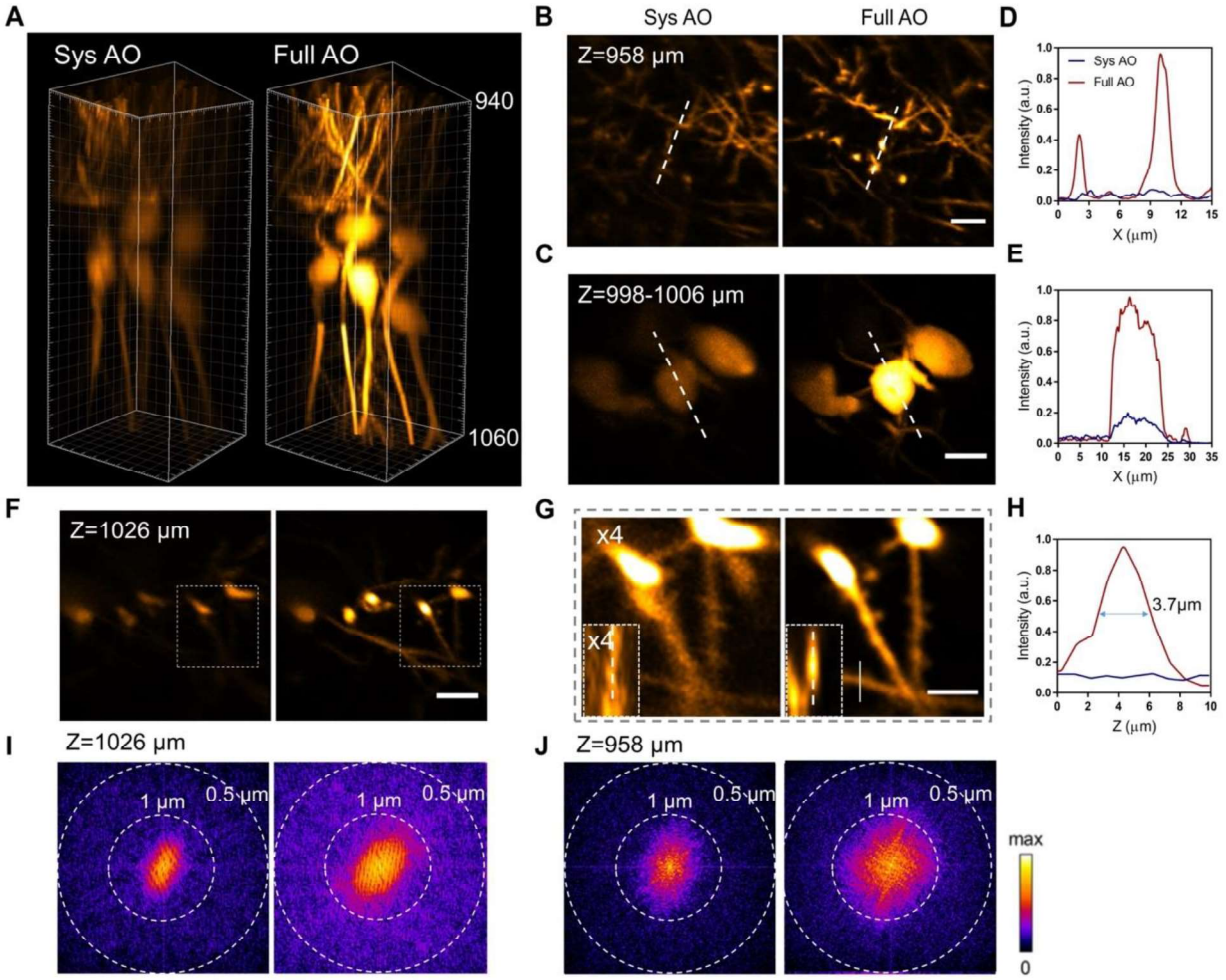


Figure S20. Pupil α-FSS-3PM improves *in vivo* imaging of neuronal structures in the hippocampus through an open skull window. (A) 3D reconstruction of hippocampal CA1 pyramidal neurons with system correction only (left) and with full correction (right). The numbers on the right side indicate the imaging depth below pia. The depth range is from 940 to 1060 μm below the pia and the lateral FOV is 50μm×50μm. (B-C) MIP of stack images from basal dendrites and soma layers, respectively, of the hippocampus in (A). (D-E) Signal profiles along the dashed lines in (B-C). (F) MIP of stack images from the layer of apical dendrites in (A). Scale bar: 10μm for (B-C) and (F). (G) Magnified views of apical dendrites from boxed regions in (F). Scale bar 5um. Insets shows the XZ view through the plane defined by the solid line in full AO image of (G). (H) Signal profiles along the dashed lines in the insets of (G), which indicates the axial resolutions in terms of FWHM. (I-J) Spectral power in spatial frequency space for the images in (F) and (B), respectively. Left: system AO; right: full AO.

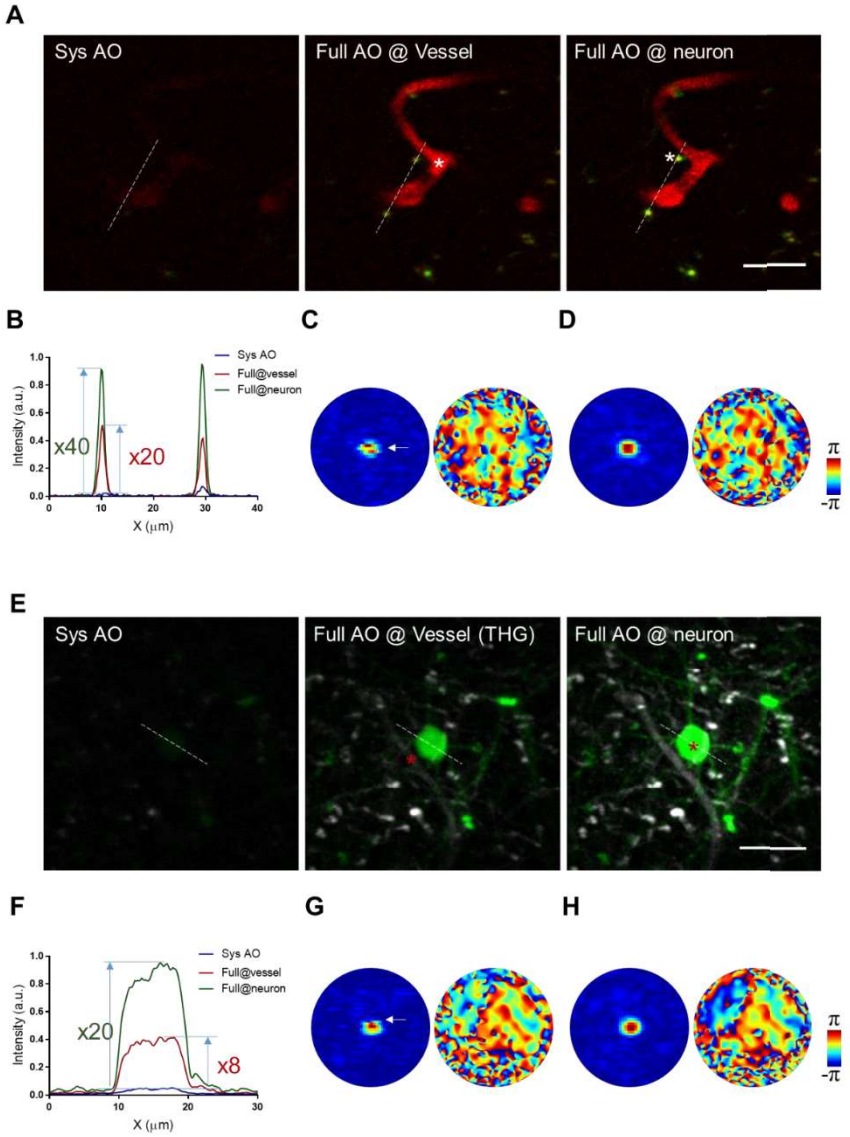
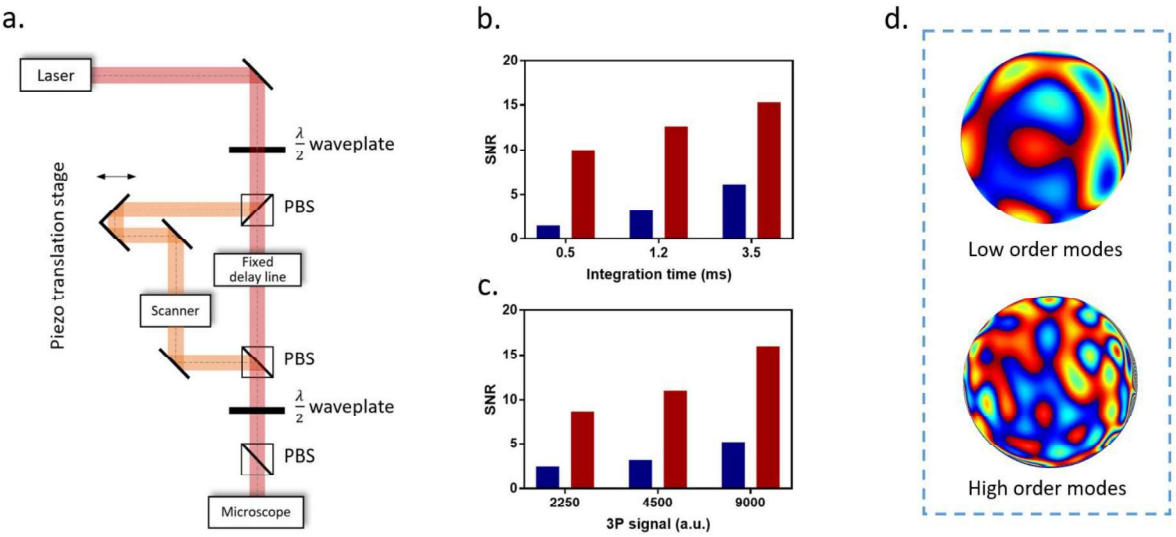
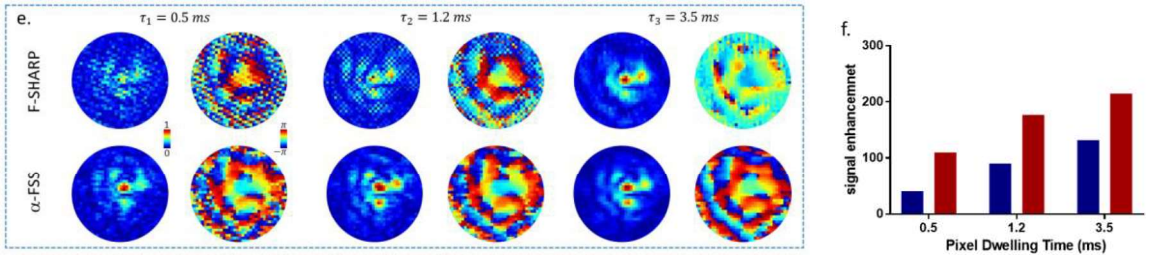


Figure S21. Direct focus sensing using fluorescence and THG signals from vessels. (A) Merged images of YFP-labeled neurons (green) and Texas-Red-labeled vessels (red) at 330μm below pia of a Thy1-YFP mouse through an intact skull with system correction only (left) and with full correction of conjugate AO by using fluorescence from vessel (middle) and neuron (right). Scale bar: 20 μm. (B) Signal profiles of green channel along the dashed lines in (A). (C) PSF magnitude and the corrective phase pattern, corresponding to focus sensing with vessel signals (indicated by the asterisk) as in the middle image of (A). The arrow indicates the artefact in PSF sensing caused by moving red blood cells in the vessel. (D) PSF magnitude and the corrective phase pattern, corresponding to focus sensing with fluorescence signals (indicated by the asterisk) as in the right image of (A). (E) Merged image of YFP-labeled neurons (green) and THG signal (white) through a thinned skull window 310-350μm below pia with system correction only (left)

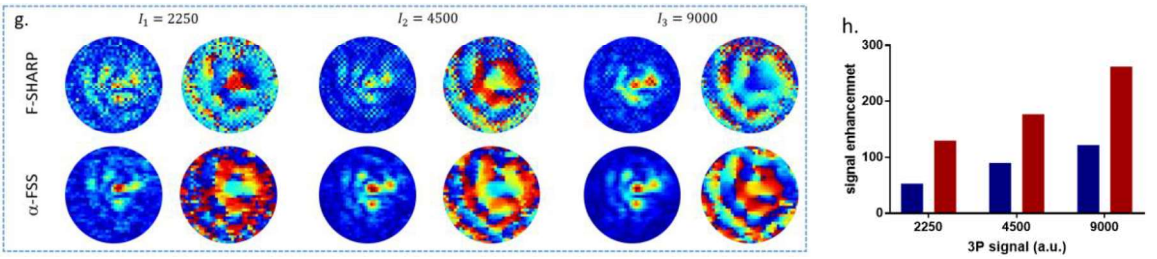
and with full correction of conjugate AO by using THG signal from vessel (middle) and YFP signal from neuron (right). Scale bar: 20 μm. (F) Signal profiles of green channel along the dashed lines in (E). (G) PSF magnitude and the corrective phase pattern, corresponding to focus sensing with vessel THG signals (indicated by the asterisk) as in the middle image of (E). The arrow indicates the artefact in PSF sensing caused by moving red blood cells. (H) PSF magnitude and the corrective phase pattern, corresponding to focus sensing with fluorescence signals (indicated by the asterisk) as in the right image of (E).


a.
Laser
λ/2 waveplate
PBS
Fixed delay line
Scanner
Piezo translation stage
PBS
λ/2 waveplate
PBS
Microscope
b.
SNR
Integration time (ms)
0.5
1.2
3.5
c.
SNR
3P signal (a.u.)
2250
4500
9000
d.
Low order modes
High order modes

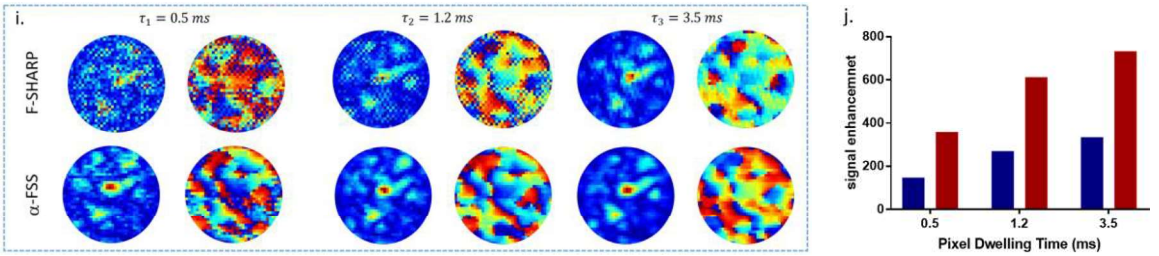
1


e.
$\tau_1 = 0.5\ ms$
$\tau_2 = 1.2\ ms$
$\tau_3 = 3.5\ ms$
F-SHARP
α-FSS
f.
signal enhancemnet
Pixel Dwelling Time (ms)

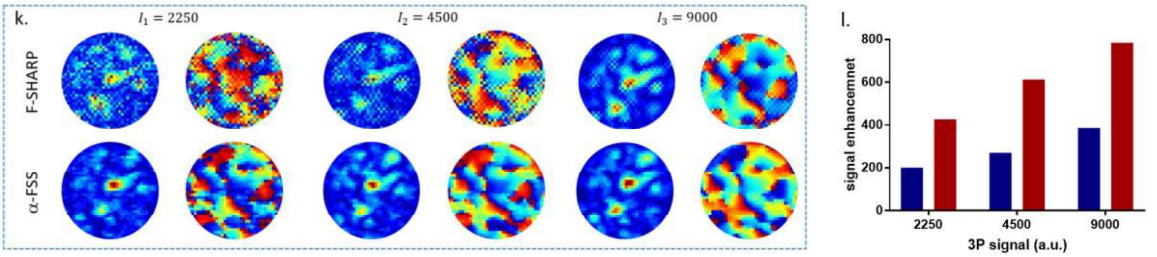
2


g.
$I_1 = 2250$
$I_2 = 4500$
$I_3 = 9000$
F-SHARP
α-FSS
h.
signal enhancemnet
3P signal (a.u.)

3


i.
$\tau_1 = 0.5\ ms$
$\tau_2 = 1.2\ ms$
$\tau_3 = 3.5\ ms$
F-SHARP
α-FSS
j.
signal enhancemnet
Pixel Dwelling Time (ms)

4


k.
$I_1 = 2250$
$I_2 = 4500$
$I_3 = 9000$
F-SHARP
α-FSS
l.
signal enhancemnet
3P signal (a.u.)

5

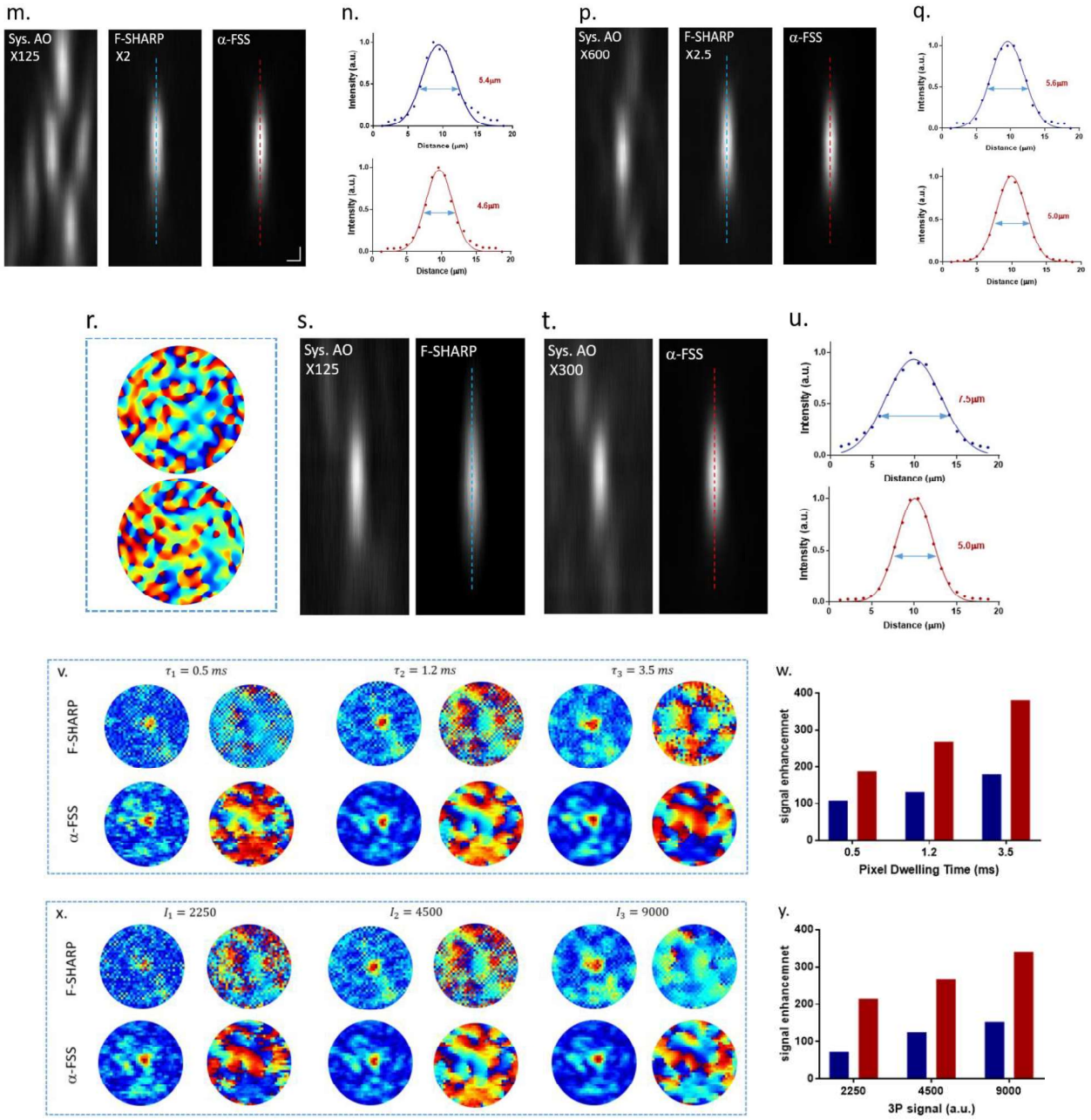


Figure. S22. Performance comparison between the phase stepping and phase modulation with lock-in detection scheme. (a) Module of phase stepping scheme for F-SHARP 3P imaging. Multiple phase difference between the stationary and scanning beams are precisely generated by a closed-loop piezo translation stage of theoretical resolution up to 5 nm (3-axis NanoMax Flexure Stages, Thorlabs Inc.). To convert the α-FSS system (Figure S1) into a F-SHARP system, only the module of phase modulation with lock-in detection is replaced by the module of phase stepping and the rest of system remain the same. In the F-SHARP experiments, the phase was being stepped after every frame using a closed-loop piezo translation stage (Papadopoulos, I. N., Jouhanneau, J.-S., Poulet, J. F. A. & Judkewitz, B. Scattering compensation by focus scanning holographic aberration probing (F-SHARP). Nat. Photonics 11, 116 (2017)). **(b)-(c)** SNR comparison between

the phase stepping and phase modulation with lock-in detection scheme under the same set of conditions (blue: signals measured from F-SHARP; red: signals measured from α-FSS). To compare the SNRs of two methods, the strong stationary beam and weak scanning beam of 10:1 intensity ratio were precisely aligned to keep overlapping with each other temporally and spatially. They were focused to an aberration free sample of Rhodamine 6G (R6G) solution. This allowed us to compare the signal quality of the E-field PSF measured by α-FSS ($E^{\text{lockin}} = X + jY$, here X, Y are in-phase and quadrature outputs of lock-in detection, respectively.) and F-SHARP based on phase stepping ($E^{\text{F-SHARP}} = (I_0 - I_\pi) + i \cdot \left(I_{\pi/2} - I_{3\pi/2}\right)$ (Papadopoulos, I. N., Jouhanneau, J.-S., Poulet, J. F. A. & Judkewitz, B. Scattering compensation by focus scanning holographic aberration probing (F-SHARP). *Nat. Photonics* **11**, 116 (2017).). Specifically, we compared the amplitude of E^{lockin} ($E^{\text{lockin}} = \sqrt{X^2 + Y^2}$) with that of $E^{\text{F-SHARP}}$ ($|E^{\text{F-SHARP}}| = \sqrt{(I_0 - I_\pi)^2 + \left(I_{\pi/2} - I_{3\pi/2}\right)^2}$). The SNRs of E^{lockin} and $E^{\text{F-SHARP}}$ were calculated by the ratios of their mean values to their standard deviations, respectively. In (b), the 3P signal was fixed at 4500 (a.u.) and signal integration time were set to 0.5 ms, 1.2 ms and 3.5 ms, respectively. For these integration time, the corresponding time constants for the lock-in amplifier were set to 0.3 ms, 1 ms, 3 ms, respectively, which are limited by the discrete choices of the time constant of the commercial lock-in amplifier. The time constants are close to the integration time for F-SHARP measurements, but shorter. This ensured that the chosen time constants would not give advantage for α-FSS in the performance comparison between two methods. In (c), the signal integration time was fixed at 1.2 ms and the 3P signal intensities were set to 2250, 4500, 9000 (a.u.), respectively, by changing the excitation power. **(d)** Phase patterns of artificially generated aberrations. To study how α-FSS and F-SHARP measure and correct the identical aberration from the same sample, an aberration-free sample of 1 μm fluorescent beads in agarose was used for α-FSS and F-SHARP 3P imaging, respectively. The aberration was artificially introduced to the excitation beam by applying the aberration phase pattern to the SLM (Figure S1). In this way, the performance of α-FSS and F-SHARP 3P imaging can be directly compared even when they are not conducted at the same time as long as the aberration introduced to the excitation is identical. Here, the artificial aberration was based on the combination of a series of Zernike modes (1. Streich, L., Boffi, J.C., Wang, L. et al. High-resolution structural and functional deep brain imaging using adaptive optics three-photon microscopy. Nat Methods 18, 1253–1258 (2021); 2. Rodríguez, C., Chen, A., Rivera, J.A. et al. An adaptive optics module for deep tissue multiphoton imaging in vivo. Nat Methods 18, 1259–1264 (2021)). Specifically, we created two aberration patterns in this study by using the combinations of low-order Zernike modes (4-21st mode) and high-order Zernike modes (4-105th mode), respectively. The first three Zernike polynomials (piston, tip, and tilt) were not included because they do not change the shape of the PSF. To avoid any bias to a specific mode, the amplitude or weight of individual Zernike mode in a combination was generated randomly. **(e)** E-field PSF amplitudes and phases measured by F-SHARP and α-FSS under different PSF pixel dwelling time when low-order aberration was introduced to the excitation (left: normalized amplitude; right: phase). The initial 3P signal intensity was fixed at 4500 (a.u.). The pixel dwelling time at 0.5 ms, 1.2 ms and 3.5 ms are corresponding to the acquisition time for a 48×48 pixel image of E-field PSF at ~1.2 s, 2.8 s and 8.1 s, respectively. **(f)** Corresponding signal gains achieved by

F-SHARP and α-FSS, respectively. (blue: F-SHARP; red: α-FSS). **(g)** E-field PSF amplitudes and phases measured by F-SHARP and α-FSS for low-order aberration at different initial 3P signal intensities before AO correction of 2250, 4500 and 9000 (a.u.), respectively, and the pixel dwelling time was fixed at 1.2 ms. **(h)** Corresponding signal gains achieved by F-SHARP and α-FSS, respectively. **(i)** E-field PSF amplitudes and phases measured by F-SHARP and α-FSS for high-order aberration at different pixel dwelling time. **(j)** Corresponding signal gains achieved by F-SHARP and α-FSS, respectively. **(k)** E-field PSF amplitudes and phases measured by F-SHARP and α-FSS for high-order aberration at different initial 3P signal intensities. **(l)** Corresponding signal gains achieved by F-SHARP and α-FSS, respectively. **(m)** XZ view of a 1μm fluorescent bead image with system correction only (left) and with F-SHARP and α-FSS correction (middle, right) for low-order aberration. **(n)** Signal profiles along the dashed lines in (m) are plotted with a curve fitted using a Gaussian function (upper: F-SHARP; lower: α-FSS). **(p)** XZ view of a bead image with system correction (left) and with F-SHARP and α-FSS correction (middle, right) for high-order aberration. **(q)** Signal profiles along the dashed lines in (p) (upper: F-SHARP; lower: α-FSS). **(r)** Correction phase patterns for a 1 μm fluorescent bead at 300 μm under an intact skull of two-month mouse (upper: F-SHARP experiments; lower: α-FSS experiments). In this study, we compared the α-FSS and F-SHARP 3P imaging of 1 μm fluorescent beads under the intact skull of two months mouse. Since α-FSS and F-SHARP 3P imaging could not be conducted at the same time, it was impossible to image the beads in an identical skull region. Therefore, the aberrations induced by the skull in α-FSS and F-SHARP 3P imaging experiments were different. The results do not provide a direct performance comparison between two methods. **(s)** XZ view of a 1μm fluorescent bead imaged at 300 μm under the intact skull with system correction only (left) and with F-FHARP correction (right). **(t)** XZ view of a 1μm fluorescent bead imaged at 300 μm under the intact skull with system correction only (left) and with α-FSS correction (right). **(u)** Signal profiles along the dashed lines in (s) and (t) (upper: F-SHARP; lower: α-FSS). **(v)** E-field PSF amplitudes and phases measured by F-SHARP and α-FSS at different pixel dwelling time. **(w)** Corresponding signal gains achieved by F-SHARP and α-FSS, respectively. **(x)** E-field PSF amplitudes and phases measured by F-SHARP and α-FSS at different initial 3P signal intensities. **(y)** Corresponding signal gains achieved by F-SHARP and α-FSS, respectively.

DISCUSSIONS:

In terms of the comparison of FOV enlargement by conjugate AO, we found no significant difference between two methods, indicating that although the quality of the corrective phase pattern measured from F-SHARP is lower, the noise may be evenly distributed over the whole phase pattern and the FOV enlargement is not quite affected. We also analyzed the lock-in detection and phase-stepping method for the phase measurement of periodical ultrashort pulse signals. Mathematically, the orthogonal X (i.e., in-phase) and Y (i.e. quadrature) outputs of an analog lock-in detection are given by:

$X = \frac{1}{T}\int_0^T I(x,t)\left(\frac{1}{2}e^{j\omega t} + \frac{1}{2}e^{-j\omega t}\right)dt$ and $Y = \frac{1}{T}\int_0^T I(x,t)\left(\frac{1}{2j}e^{j\omega t} - \frac{1}{2j}e^{-j\omega t}\right)dt$, respectively.

Therefore, the E-field PSF of DFSS is calculated as: $E^{\text{lockin}} = X + jY \propto \frac{1}{T}\int I(x,t)e^{j\omega t}dt$

Because of ultrashort pulse excitation, the 3P fluorescence signal $I(x,t)$ can be approximately written as a periodic sequence of pulse function and the E-field PSF E^{lockin} becomes:

$$E^{\text{lockin}} \propto \frac{1}{T}\int \sum_{n=0}^{N-1} \delta\left(t - \frac{n}{f_r}\right) I^{\left(\frac{\omega n}{f_r}\right)} e^{j\omega t}dt = \frac{1}{T}\sum_{n=0}^{N-1} I^{\left(\frac{\omega n}{f_r}\right)} e^{\frac{j\omega n}{f_r}} \qquad (1)$$

Where T is the time constant, f_r is the laser repetition rate, $\omega = 2\pi f_m$ with f_m as the phase modulation frequency, $N = T * f_r$ is the number of laser pulses in the time window of lock-in detection, and $\frac{\omega n}{f_r}$ is the phase delay of the two beams.

For a general *K*-step phase stepping, it involves the application of a circular phase sequence $\phi_k = \frac{2\pi k}{K}$, where k is incremented from 0 to *K*-1, and the interference term is given as follow (Jerome Mertz's Introduction to Optical Microscopy, 2019, page 226, Equation 11.45):

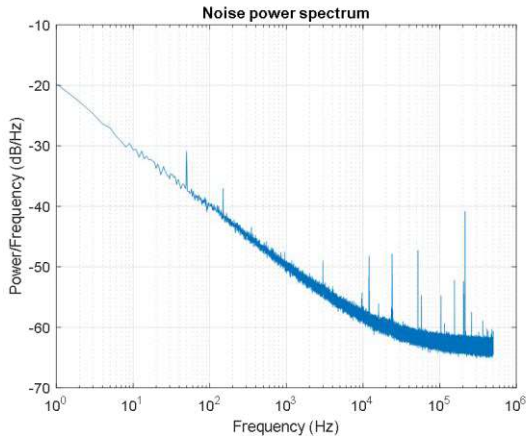
$$E^{\text{Phase Stepping}} \propto \frac{1}{K}\sum_{k=0}^{K-1} I^{(\phi_k)} e^{(j\phi_k)} \qquad (2)$$

As can be seen from Eq. (1) and (2), a general K-step phase stepping is different from the lock-in detection. However, if the K-step phase shifting is repeated over $M = N/K$ cycles, where $K = \frac{f_r}{f_m} * W$ and W is the minimal positive integer such that K is an integer, the averaged output will become:

$$E^{\text{K-step + average}} = \frac{1}{M}\sum_{m=0}^{M-1} E^{\text{K-step}} \propto \frac{1}{T}\sum_{m=0}^{M-1}\sum_{k=0}^{K-1} I^{\left(\frac{2\pi k}{K}\right)} e^{\left(j\frac{2\pi k}{K}\right)} = \frac{1}{T}\sum_{n=0}^{N-1} I^{\left(\frac{2\pi f_m n}{f_r}\right)} e^{\frac{j2\pi f_m n}{f_r}} \qquad (3)$$

The Eq. (3) is equivalent to Eq. (1), proving that a phase stepping with multiple cycle averaging will work as a digital lock-in detection and its output is equivalent to an analog lock-in detection in α-FSS when the number of measurement cycles and modulation frequency are the same. This indicates that theoretically, F-SHARP may produce the same results if K-step phase stepping is repeated over $M = N/K$ cycles with the same modulation frequency of lock-in detection.

We measured the noise power spectrum of the femtosecond laser for 3P excitation and the result is presented in the figure below. As can be seen, unlike F-SHARP, α-FSS can push the frequency of phase-sensitive detection to the MHz range, which benefits from higher signal-noise-ratio due to 1/f trend of laser noise spectrum.



Furthermore, the phase sensitive detection system based on a single-board all analog lock-in amplifier will be more advantageous for even higher frequency modulation detection when an excitation source of high repetition rate, typically at 80 MHz, is used, because the E-field PSF can be directly calculated from its analog in-phase and quadrature outputs (Saar BG, Freudiger CW, Reichman J, Stanley CM, Holtom GR, Xie XS. Video-rate molecular imaging in vivo with stimulated Raman scattering. Science. 2010 Dec 3;330(6009):1368-70. doi: 10.1126/science.1197236.). This is critical to reduce the motion effect during *in vivo* imaging.

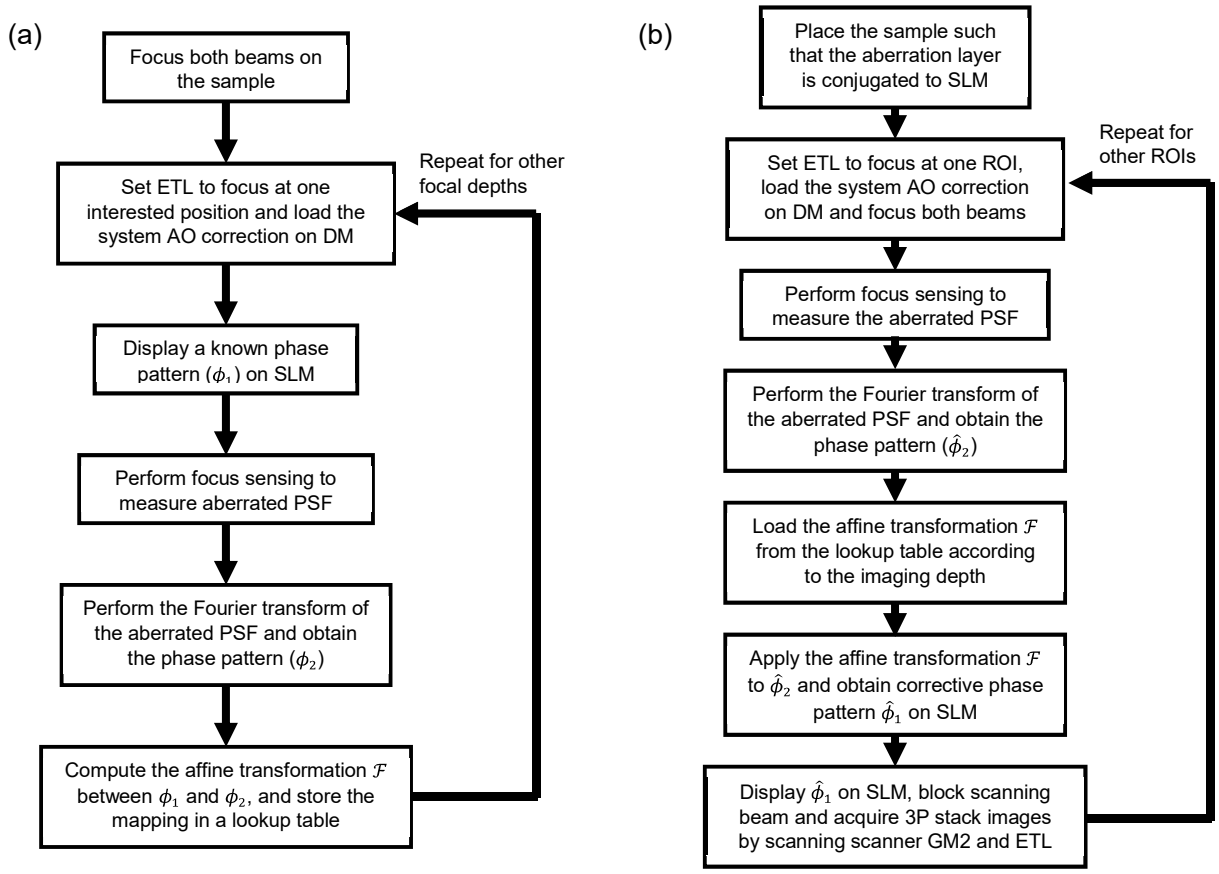


Figure S23 Flowchart diagrams for (a) calibrating the affine transformation of wavefront on SLM and objective back aperture and (b) performing α-FSS *in vivo*.

Table S1. α-FSS and 3P imaging parameters.

Figure. No.	Fluorescent labels	Window type/ skull thickness (μm)	Imaging depth (μm)	Post-objective power (mW)	Pixel dwell time (μs)	Number of frames averaged per slice	α-FSS time (s) and number of iterations
Fig. 1B, F-G	Red beads	Intact/ 100	306	2	2	4	3.5+3.5
Fig. 1E Fig. S3B	Red beads	Intact/ 100	310	2	2	16	3.5+3.5
Fig. 1F Fig. S3C	Red beads	Intact/ 100	306	3	2	4	3.5+3.5
Fig. 2 Fig. S6-7	YFP/ Texas red	Intact/ 100	0-360	15	1	32	3.5+3.5
			360-540	20-40	1	32	3.5+3.5+3.5
			540-800	40-75	1	32	3.5
Fig. 3A	GCaMP6s	Intact/ 110	315	40	1	-	3.5+3.5+3.5
Fig. 3B	GCaMP6s	Intact/ 110	350	36	1	-	3.5+3.5
Fig. 3C	GCaMP6s	Intact/ 110	445	60	1	-	3.5
Fig. 4A	GFP	Intact/ 100	158-182 268-292 378-402	40-80	1	13	3.5+3.5
Fig. 4D	MeOX4/ GFP	Intact/ 140	256-290	60	1	48	3.5+3.5+3.5
Fig. 5A-B	GFP	Open	882-902	17	1	64	1.5+1.5
Fig. 5E-F	GFP	Open	984-990	35	1	128	1.5+1.5
Fig. 5I-J	GFP	Open	1136-1140	64	1	128	4.5+1.5
Fig. S2	Red beads	Intact/ 100	310	1.4	2	16	3.5+3.5+3.5
Fig. S4A	GFP	Intact/ 100	280-320	30	1	32	4.5+3.5
Fig. S4B	GFP	Intact/ 100	280-320	50	1	32	3.5+3.5
Fig. S5A	GFP	Intact/ 110	282-292	45	2	32	3.5+3.5+3.5
Fig. S5E	GFP	Intact/ 110	285-315	40	2	32	3.5+3.5+3.5

Fig. S6A	GFP	Intact/ 100	140-480	20-80	1	32	3.5+3.5
Fig. S7A	GFP	Intact/ 100	480-500	100	1	64	3.5+3.5
Fig. S10A	YFP	Intact/ 100	100-110	15	1	32	3.5+3.5
			290-310	20	1	64	3.5+3.5
			520-530	45	1	64	3.5+3.5+3.5
			682-722	80	1	64	3.5+3.5+3.5
Fig. S11A-D	Red beads	Intact/ 100	300	3	1	16	3.5
Fig. S12A	GCaMP6s	Intact/ 110	170-310	20-50	2	-	3.5+3.5+3.5
Fig. S13	YFP	Intact/ 140	0-350	30	1	32	3.5+3.5+3.5
			350-450	40-75	1	32	3.5+3.5
			450-550	75	1	64	3.5+3.5
Fig. S14A-B	YFP	Thinned/ 50	0-440	4-8	1	32	3.5+3.5
			440-740	8-20	1	32	3.5+3.5+3.5
Fig. S14I	YFP	Thinned/ 50	660-690	60	1	32	3.5+3.5
Fig S15	YFP	Thinned/ 50	124-128	5	1	32	3.5+3.5
			302-306	8	1	32	3.5+3.5
			520-530	20	1	32	3.5+3.5+3.5
			670-680	60	1	32	3.5+3.5
Fig. S16A	GCaMP6s	Thinned/ 50	425	25	1	-	3.5+3.5+3.5
Fig. S16C	GCaMP6s	Thinned/ 50	585	60	1	-	3.5+3.5
Fig. S17A	GCaMP6s	Thinned/ 50	395 480 590	24-60	2	-	3.5+3.5 +3.5+3.5
Fig. S18A	GFP	Open	780	12	1	64	1.5+1.5
Fig. S19A	GFP	Open	940-980	48	1	128	1.5+1.5
			980-1020	35	1	128	1.5+1.5
			1020-1040	48	1	128	1.5+1.5
Fig. S20A	YFP/ Texas Red	Intact /100	330	30	1	16	3.5+3.5+3.5
Fig. S20E	YFP/ THG	Thinned/ 50	310-350	15	1	32	3.5+3.5+3.5

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Movie S1. Single correction of conjugate AO recovers subcellular resolution over a large volume (z = 100-500 µm) for *in vivo* imaging of microglia through the intact skull.

Movie S2. AO-assisted *in vivo* multi-plane calcium imaging of neuronal activity through the intact skull.

Movie S3. AO-assisted precise laser microsurgery and time-lapse imaging of microglial dynamics through the intact skull.